

Delta Dependency Prioritizes Candidate Paralog Dependencies Across Solid Tumor Types

Qingqing Mo^{1,2}, Tao Zhu^{1,2,*}

¹ Department of Obstetrics and Gynecology, National Clinical Research Center for Obstetrics and Gynecology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, 1095 Jiefang Avenue, Wuhan 430030, China

² Key Laboratory of Cancer Invasion and Metastasis (Ministry of Education), Hubei Key Laboratory of Tumor Invasion and Metastasis, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, 1095 Jiefang Avenue, Wuhan 430030, China

* Correspondence: zhutao@tjh.tjmu.edu.cn

Abstract

Background: Paralog compensation is a specific synthetic-lethal (SL) mechanism: when a tumor-suppressor driver is lost to mutation, its sequence-similar paralog can become conditionally essential. Systematic prioritization of paralog-SL candidates remains limited; existing models integrate multiple features, whereas a measurable dependency-shift statistic offers a complementary, transparent strategy.

Results: We introduce Delta Dependency (DD), an interpretable metric measuring the dependency shift on a paralog between driver-mutant and wild-type cell lines in DepMap (1,208 lines, 23 solid tumor types). On twelve curated, evidence-tiered pairs in the primary literature-derived benchmark (three gynecological lineages), signed DD achieved AUROC = 0.629 in a positive-unlabeled, internal-consistency evaluation (pair-clustered bootstrap 95% CI 0.253–0.933; 0.613 excluding two DepMap-labelled pairs; 0.525 for the eight pairs with pre-DepMap experimental evidence; the sole lineage-evaluable Tier A/B pair, ARID1A→ARID1B, ranked above all unlabeled controls in both evaluable lineages). Exploratory analysis of seven CPTAC cohorts ($n = 771$) detected protein-level paralog co-variation. Dependency-window scoring recovered the established ARID1A→ARID1B pair (DWS = 2.82) as the highest-selectivity established candidate and nominated KMT2D→KMT2C as the highest-ranked non-benchmark candidate for experimental follow-up.

Conclusions: DD offers a mechanistically transparent complement to multi-feature SL prediction: each nomination traces to a measured dependency shift. DD is an association-based prioritization metric, not a causal test of synthetic lethality; all candidates require

experimental validation. An open-source R package (**paralogSL**) and reproducible pipeline are provided.

Keywords: Synthetic lethality, paralog compensation, DepMap, CPTAC proteomics, Delta Dependency, dependency window

Background

The PARP inhibitor story crystallizes what makes synthetic lethality (SL) so compelling: BRCA1/2-mutant tumors depend on PARP-mediated repair, and drugging PARP kills them while sparing normal cells [1–5]. The logic is elegant; repeating it for other mutations has proven hard [6]. Available SL catalogues contain tens of thousands of heterogeneous, unevenly validated interactions [7, 55], and the existing computational prediction methods (graph neural networks [8, 9], graph convolutional networks with dual dropout [10], graph-regularized matrix factorization [11], knowledge-graph neural networks [12], and negative-sample-free contrastive learning [13]) train almost entirely on these limited known pairs. Feng et al. [14] showed that these methods degrade sharply from the CV1 setting (AUROC > 0.90 for the best methods) to the CV3 gene-pair isolation setting, where most fall to near-random performance (AUROC $\approx$ 0.5) and the best published result reaches 0.790 [15], precisely the setting where predictive models matter: nominating genuinely new SL pairs.

Paralog compensation provides a simpler route into this problem [16, 17]. Large-scale analyses of CRISPR dependency data have since established paralog pairs as the most enriched class of synthetic-lethal interactions and shown that interpretable models built on protein-interaction and evolutionary-conservation features can predict them [50]; combinatorial knockout screens have begun to map paralog synthetic lethality directly [32, 52–54, 64, 65]. Gene duplication events leave behind paralogs: sequence-similar gene pairs with overlapping biochemical roles. When a paralog acting as a tumor suppressor (the “driver”) is inactivated by mutation, its partner paralog can become conditionally essential. This is sequence-paralog SL: synthetic lethality between two genes that share high protein sequence identity (operationalized with a validated k -mer identity-enrichment proxy; Methods) and conserved functional domains. Known examples include SMARCA4→SMARCA2 (SWI/SNF ATPase subunits) [18], ARID1A→ARID1B (BAF complex subunits) [19], AKT1→AKT2 [45], and CDK4→CDK6 and MAP2K1→MAP2K2 [52]. Other catalogued pairs carry indirect or direction-restricted evidence: for EP300 and CREBBP (histone acetyltransferases), direct experiments established the reciprocal CREBBP→EP300

direction [23, 48]; for PIK3CA→PIK3CB (PI3K catalytic isoforms), the primary evidence supports PTEN→PIK3CB [25]; and CCNE1→CCNE2 is supported by developmental redundancy in mouse double knockouts [46]. Because the evidence behind catalogued pairs varies in both type and direction, we verified each primary citation and assigned every pair an evidence tier (Supplementary Table S3; Results; CCNE1→CCNE2, for example, is not evaluable under our mutation rules because CCNE1 is amplification-driven and carries no qualifying hotspot mutations). In every case, the driver and paralog share sequence homology and biochemical function; when the driver is inactivated by mutation, the remaining paralog becomes the cancer cell’s dependency.

Two related phenomena need to be separated from sequence-paralog SL. Functional analogs such as BRCA1 and BRCA2 share pathway roles (homologous recombination repair) but have no sequence homology. Pathway-level SL interactions, such as BRCA1↔PARP1 and PTEN↔PIK3CB [24, 25], exploit indirect pathway wiring rather than direct biochemical redundancy. Our analysis concentrates on sequence-paralog pairs; functional analogs including BRCA1/BRCA2 are retained as mechanistic comparators (specificity references, not verified negatives) and flagged throughout (Supplementary Table S3).

D’Antonio et al. published the first systematic analysis of negative genetic interactions between functional paralogs in cancer cell lines, identifying buffering relationships at the gene-family level [17]. That study opened the field but did not benchmark paralog-based predictions against published SL methods, did not incorporate proteomic validation, and did not address clinical stratification or therapeutic window assessment. Two practical gaps remain. First, complex multi-feature models are less directly traceable to a single experimentally measurable effect: when a neural network predicts SL, it is difficult to reconstruct why, which hinders rational experimental design. Second, moving from a ranked list to an experimentally prioritized candidate set requires conservative integration of orthogonal evidence layers; the optimal readout for paralog compensation is itself unsettled, since RNA-level and protein-level signals can diverge [33], and how protein-level compensation relates to dependency-based prioritization remains unexplored.

Here we introduce Delta Dependency (DD): the difference in mean Chronos gene-effect scores between driver-mutant and wild-type cell lines. DD complements multi-feature classifiers with a single subtraction: $|DD| = 0.386$ for ARID1A→ARID1B in ovarian cancer cells means the paralog’s essentiality score shifts by 0.386 units when the driver is mutant, a measurable quantity

that can guide experimental design. We evaluate DD across 23 solid tumor types using a 40-gene driver panel (Supplementary Table S4) curated from TCGA PanCanAtlas driver analyses [26] and the COSMIC Cancer Gene Census [49], and we support the genomic predictions with CPTAC proteomics (seven cohorts), PRISM drug screening (1,482 compounds), MSI-aware patient stratification, mutation-type analysis, dependency-window quantification, and exploratory structure-derived targetability descriptors. The primary objective was to assess whether signed DD enriches literature-supported mutation-conditioned paralog dependencies relative to unlabeled paralog pairs; secondary analyses evaluated robustness and selectivity, and the proteomic, pharmacologic, clinical, and sequence-derived analyses were exploratory.

Results

DD defines a dependency-shift metric for paralog-SL prioritization

We defined Delta Dependency as:

$$DD(D, P, c) = \bar{G}_{P, D-WT, c} - \bar{G}_{P, D-MUT, c} \quad (1)$$

where $\bar{G}$ denotes the mean Chronos gene-effect score, D is the driver gene, P the candidate paralog, and c the cancer lineage. Because lower Chronos gene-effect scores indicate stronger dependency, positive DD indicates increased dependency on paralog P in driver-mutant cell lines relative to wild-type within lineage c , consistent with paralog compensation. Throughout, DD is used as a signed quantity: only positive shifts (increased dependency in driver-mutant cells) are treated as compensation signal, and reverse-direction shifts score against it; analyses scored by $|DD|$, which also credit reverse-direction shifts, are reported as sensitivity analyses.

DD was computed for 66,595 HGNC paralog pairs across 1,208 DepMap cell lines; benchmark evaluation on the curated evaluation set (Results, benchmark comparison) showed signed DD exceeding AUROC = 0.7 in 1 of 8 evaluable lineages on the primary frame (Fig. 1a; Supplementary Table S1): Colorectal (AUROC = 0.775, $n = 62$ pairs) led, followed by Esophagogastric (0.674, $n = 50$ pairs), non-small cell lung cancer (NSCLC; 0.664), small cell lung cancer (SCLC; 0.651), Ovarian (0.611), Bladder Urothelial (0.531), Endometrial (0.519), and Breast (0.150; per-lineage cell-line counts, tested pairs, and DD AUROC values in Supplementary Fig. S1). Under the relaxed ≥ 3 -per-group sensitivity frame (12 evaluable lineages), 6 of 12 exceeded

0.7, with Biliary Tract (0.990), Pancreatic (0.962), and Melanoma (0.840) the strongest performers. Scoring by $|\text{DD}|$, which also credits reverse-direction shifts, yields higher values (7 of 8 primary-frame lineages above 0.7) but is reserved for sensitivity analysis because reverse-direction shifts are not paralog compensation (Methods). Performance did not differ significantly by oncogenic mechanism (Fig. 1b; sensitivity frame, because the primary frame retains only one oncogene-driven lineage): TSG-driven cancers had a mean AUROC of 0.649 ($n = 9$ lineages) versus 0.834 for oncogene-driven cancers ($n = 3$ lineages; permutation $p = 0.127$, exact Mann-Whitney $p = 0.145$). With only three oncogene-driven lineages the comparison is severely underpowered, and we report it as descriptive.

We compared DD against eight published SL prediction methods in the CV3 setting reported by Feng et al. (2024) [14] (Supplementary Table S12). DD achieved AUROC = 0.629 without any training, below the best published result in that setting (SLMGAE [15], 0.790), though we stress that the evaluation frameworks are analogous, not identical. We therefore treat the published values as contextual reference points rather than a formal benchmark. Restricting DD to the three pairs with $\geq 30\%$ sequence identity (two positives) yielded AUROC = 1.000; with so few pairs this estimate is anecdotal and is reported as an exploratory sensitivity analysis without a significance claim. For a true head-to-head comparison on the same 72 paralog pairs (6 known positives), we trained four standard classifiers, namely logistic regression (LR), random forest (RF), SVM-RBF, and SVM-Linear, on five features ($|\text{DD}|$, the paralog compensation score ($\text{PCS} = \max(\Delta\text{Expression}, 0) \times \max(\text{necessity}, 0)$, where $\text{necessity} = -\overline{\text{Chronos}}$ so that larger values mark more essential paralogs), $|\Delta\text{Expression}|$, necessity, and mutation frequency; the composite score was excluded from classifier inputs because it is a deterministic function of the other features) under leave-one-pair-out cross-validation. The composite score (AUROC = 0.831) and SVM-RBF (0.841) performed best, with no detectable difference between them at this sample size (paired-bootstrap $\Delta\text{AUROC} = -0.01$, 95% CI -0.21 to $+0.18$; the interval is too wide to establish equivalence), followed by random forest (0.722) and DD alone (0.672 on this mean-aggregated pair frame; see “Evaluation frameworks”); the linear classifiers were unstable (SVM-Linear 0.240, LR 0.136). With only 6 positives, multi-feature training amplifies noise and no classifier result should be over-interpreted. On the same 72-pair frame, the composite score reached AUPRC = 0.356 ($4.3\times$ the baseline prevalence of 0.083), confirming enrichment beyond random expectation. The composite is a deterministic weighted combination of min-max-normalized PCS (0.50), $|\text{DD}|$ (0.20), $-\log_{10}(q)$ (0.15), and mutation frequency (0.15),

with weights fixed in the analysis code (`pcs.py`) and not fitted by nested resampling; because the same benchmark informed its development, its AUROC is descriptive context rather than independent validation. Min-max scaling of the composite components is computed within each cross-validation fold on training pairs only; this fold-internal scaling leaves the composite AUROC unchanged to three decimals (0.831) and changes AUPRC from 0.356 to 0.368. The composite and the dependency-window score use magnitude-based $|\text{DD}|$ as an effect-size term, whereas signed DD remains the directional metric for biological interpretation. These comparisons are confined to the paralog-based SL task; DD is not intended for pathway-level interactions such as BRCA1 $\leftrightarrow$ PARP1. A DD + k -mer identity-enrichment filter ($\geq 30\%$ proxy) achieved AUROC = 1.000 on a high-identity subset of 3 pairs (2 known SL), but with only two positives this is anecdotal and we report it without a significance claim. Unlike neural networks, every DD nomination maps to a single measured quantity, the dependency shift, which makes it straightforward to design a validation experiment.

Component decomposition on the same test set: signed DD (AUROC = 0.629) versus PCS (0.825), Δ Expression alone (0.547), and necessity (0.642; Fig. 1d). A paired bootstrap over entries found no significant difference between DD and any other component (PCS minus DD +0.195, 95% CI -0.132 to +0.545). Of the components, PCS carries substantial signal, which motivates the composite score.

We evaluated DD on evidence-tiered test sets (Table S3), reclassifying the twelve curated pairs after verifying each primary citation. Three pairs have direct genetic synthetic-lethal evidence from dual-gene perturbation assays (Tier A: AKT1 $\rightarrow$ AKT2 [45], CDK4 $\rightarrow$ CDK6 and MAP2K1 $\rightarrow$ MAP2K2 [52]). Two further pairs are demonstrated selective dependencies in driver-mutant cells with functional validation (Tier B: SMARCA4 $\rightarrow$ SMARCA2 [18], ARID1A $\rightarrow$ ARID1B [19]); the Tier A $\cup$ Tier B set constitutes the primary literature-derived benchmark. Five pairs rest on indirect, reciprocal-direction-only, or DepMap-derived evidence (Tier C: EP300 $\rightarrow$ CREBBP [23, 48], PIK3CA $\rightarrow$ PIK3CB [25], CCNE1 $\rightarrow$ CCNE2 [46], FBXW7 $\rightarrow$ FBXW2, PPP2R1A $\rightarrow$ PPP2R1B), and two serve as mechanistic comparators (BRCA1 $\leftrightarrow$ BRCA2, STK11 $\rightarrow$ SIK1). On the full curated set of twelve pairs, evaluated in the three gynecological lineages carried through the benchmark pipeline (Ovarian, Endometrial, Cervical), signed DD achieved AUROC = 0.629 (110 driver $\times$ paralog $\times$ lineage entries, 8 positive entries). Label-quality sensitivity analyses: excluding the two DepMap-derived pairs gave AUROC = 0.613, restricting to the eight pairs with pre-DepMap experimental evidence gave 0.525, and a direction-strict analysis relabelling the

EP300→CREBBP entries as non-positive left the estimate unchanged at 0.629. The signed-metric estimate thus attenuates when DepMap-era labels are removed, and we report this dependence explicitly (Discussion). On the primary Tier A $\cup$ Tier B frame, both evaluable positive entries (the same pair, ARID1A→ARID1B, in Endometrial and Ovarian cancers) ranked above all 102 unlabeled control entries (AUROC = 1.000); SMARCA4→SMARCA2 and the three Tier A pairs had too few qualifying driver-mutant cell lines for lineage-level evaluation under the ≥ 5 -per-group rule. With at most 2–3 strictly validated positives, AUROC computed on small subsets is not informative, so we rely on the full-set estimate and supplement it with effect-size ranking: ARID1A→ARID1B has the largest lineage-level DD (0.386 in Ovarian cancer; Hedges’ $g = 1.39$; Supplementary Table S7) and a top-tier dependency-window and selectivity profile (Supplementary Table S9), providing orthogonal prioritization beyond classification performance.

Leave-one-lineage-out AUROC over the three evaluation lineages was stable (range 0.606–0.692). Bootstrap resampling of pairs (1,000 iterations, 72 pairs, 6 positives) on the max-aggregated pair frame gave a 95% CI of 0.294–0.845 (Supplementary Fig. S2). On the primary lineage-level frame, a pair-clustered bootstrap (10,000 iterations, resampling whole pairs with all their lineage entries) gave a wider 95% CI of 0.253–0.933 around the 0.629 estimate, a driver-block bootstrap (10,000 iterations, resampling driver genes together with all their pairs) gave a 95% CI of 0.272–0.929, and pair-level label permutation (10,000 permutations) gave $p = 0.122$; the effective sample size is ≈ 85 of the 110 entries after accounting for within-pair correlation. In the pair-level aggregated frameworks, null distributions from 10,000 label permutations were centered at 0.501 and neither aggregated estimate significantly exceeded its null (max aggregation 0.598, empirical $p = 0.223$; mean aggregation 0.672, $p = 0.090$; Supplementary Fig. S2). With only 8 positive entries (6 positive pairs) the evaluation is underpowered (at the observed 0.629, power to reject a null of 0.5 is 0.34, and ≥ 45 validated positive pairs would be required for 80% power at $\alpha = 0.05$; see Methods), so all benchmark AUROCs are reported as point estimates with explicit uncertainty, with the lineage-level frame retained as the primary framework throughout. Copy number alterations explained $< 10\%$ of gene-effect variance for all 23 paralog genes tested (maximum $R^2 = 0.057$; Supplementary Fig. S3), and multivariate regression controlling for CNV left DD estimates virtually unchanged (mean $|\Delta DD| = 0.005$); because Chronos scores already model out screen-level copy-number bias, this adjustment targets residual paralog-specific confounding. ARID1A→ARID1B remained highly significant after

213 CNV adjustment (adjusted $p = 6.5 \times 10^{-42}$), as did SMARCA4→SMARCA2 ($p = 3.5 \times 10^{-29}$)
 214 and EP300→CREBBP ($p = 1.4 \times 10^{-13}$). Controlling for expression ($G_P \sim D_{\text{mutation}} + \text{Expr}_P$)
 215 produced similar results (mean $|\text{ADD}| = 0.014$); significance was retained by the strongest pairs
 216 (ARID1A→ARID1B $p = 4.1 \times 10^{-41}$, PIK3CA→PIK3CB $p = 1.5 \times 10^{-11}$, EP300→CREBBP $p =$
 217 6.3×10^{-13}), whereas weaker pairs did not reach significance in the pan-cancer frame, consistent
 218 with lineage-specific compensation. TP53 co-mutation, present in 76/121 ARID1A-mutant lines,
 219 had negligible impact on ARID1A→ARID1B ($\text{ADD} = 0.0004$, adjusted $p = 5.8 \times 10^{-42}$). Adding
 220 lineage fixed effects attenuated but preserved the association (lineage-adjusted $p = 2.2 \times 10^{-13}$;
 221 TP53- and CNV-adjusted with lineage fixed effects $p = 7.0 \times 10^{-13}$). Re-estimating these mod-
 222 els with standard errors clustered by Oncotree lineage likewise preserved significance for all
 223 three headline pairs (ARID1A→ARID1B $p = 1.7 \times 10^{-9}$; SMARCA4→SMARCA2 $p = 5.2 \times 10^{-3}$;
 224 EP300→CREBBP $p = 3.2 \times 10^{-6}$; Supplementary Table S11).

225 External digenic screens. As a fully external test we applied the frozen pipeline to two recent
 226 combinatorial CRISPR screens that played no role in method development, feature choice, or
 227 candidate selection [64, 65]: the directed screen of Harle et al. (472 nominated pairs across
 228 27 melanoma, lung, and pancreatic cell lines; 117 experimentally determined synthetic-lethal
 229 pairs) and the unbiased paralogome-wide screen of Flister et al. (36,648 pairs screened in NCI-
 230 H1299 and MDA-MB-231; 146 lethal pairs). Scoring each pair by mutation-agnostic $\max|\text{DD}|$
 231 (Methods), dependency displacement distinguished screen hits from non-hits in both datasets
 232 (Harle: AUROC = 0.619, permutation $p = 0.0002$, 411 evaluable pairs including 98 hits; Flister:
 233 AUROC = 0.696, $p = 0.0001$, 29,457 evaluable pairs including 114 hits; per model 0.671 and
 234 0.744; Additional file 1, Supplementary Methods). Dependency displacement therefore carries
 235 reproducible signal for digenic synthetic lethality at scale, on experimental platforms entirely
 236 independent of DepMap; the null in4mer comparison (Limitations) reflects its 10-positive sample
 237 size rather than an absence of signal. Two qualifications apply. First, the Harle library was
 238 nominated from prior RNAi and digenic studies and is therefore partially enriched for previously
 239 reported candidates; the Flister screen is free of this enrichment. Second, the driver-conditioned,
 240 directional proposition that defines DD could not be adjudicated in either screen, because only
 241 three curated pairs per dataset had any driver-mutant models: pooled digenic hit rates were
 242 0/9 in driver-mutant versus 27/121 in wild-type Harle models (Fisher $p = 1.0$) and 2/6 versus
 243 59/235 in Flister models ($p = 0.475$). This reflects the rarity of our driver mutations among
 244 screened lines and the low genotype penetrance of digenic interactions reported by both studies.

245 The single directionally consistent case, EP300→CREBBP in Flister (hits in 2/3 mutant versus
246 20/46 wild-type models), is anecdotal.

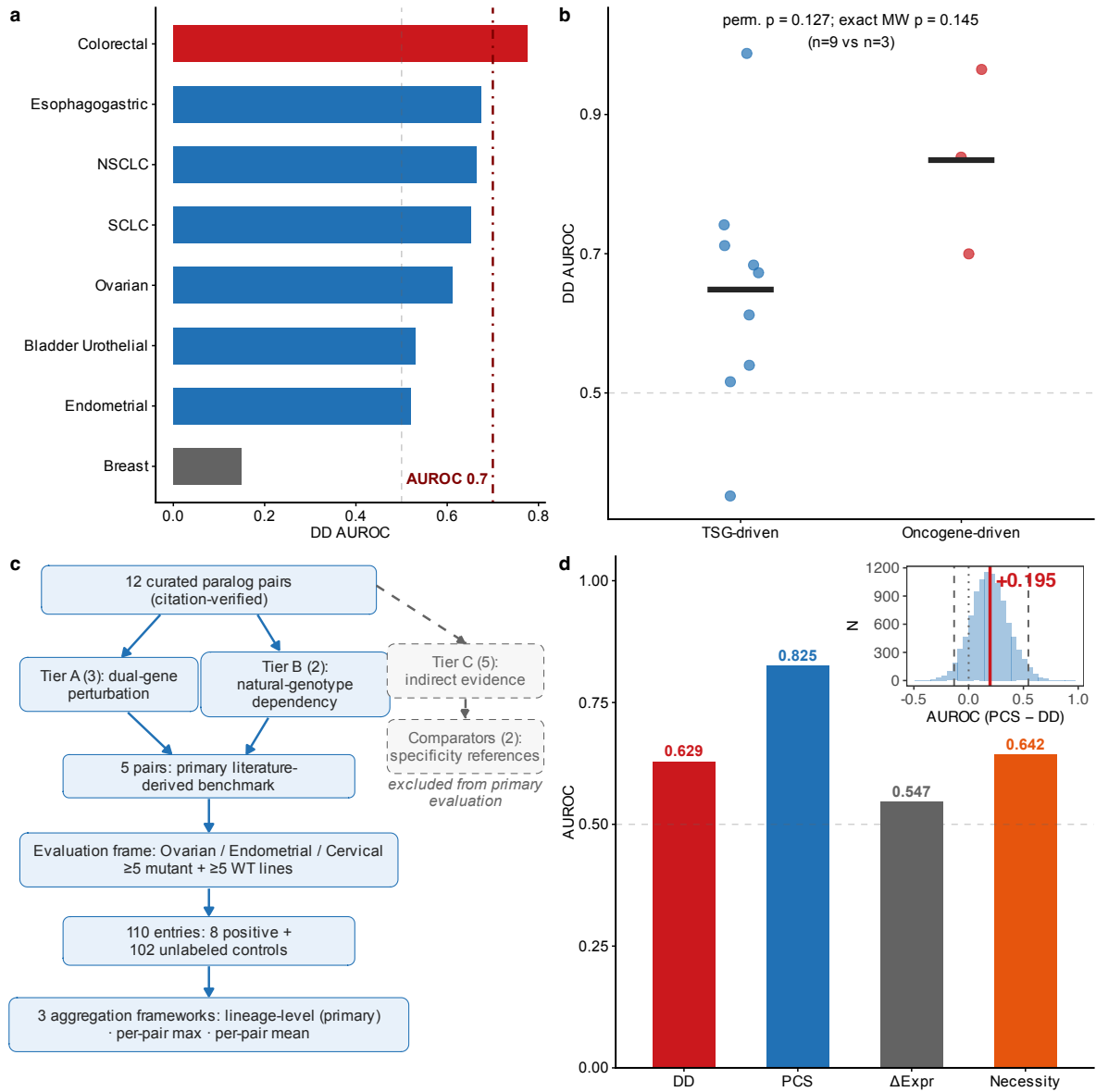


Fig. 1: Delta Dependency: pan-cancer performance, benchmark, and validation. **a**, DD AUROC across 8 evaluable solid tumor types on the primary ≥ 5 -per-group frame (red: AUROC > 0.7 ; blue: 0.5 – 0.7 ; gray: < 0.5). Dashed line, random classifier (0.5). Bar labels show n tested pairs. Sample sizes and full lineage names in Supplementary Table S1; the ≥ 3 -per-group sensitivity frame yields 12 evaluable lineages. **b**, DD performance stratified by oncogenic mechanism. Mean AUROC does not differ significantly between TSG-driven tumors ($n = 9$; mean 0.649) and oncogene-driven tumors ($n = 3$; mean 0.834 ; permutation $p = 0.127$). Points, individual lineages; center line, group mean. Computed on the ≥ 3 -per-group sensitivity frame because the primary ≥ 5 frame retains only one oncogene-driven lineage. **c**, Benchmark selection and evaluation framework. Twelve curated pairs were evidence-tiered after verification of each primary citation (three Tier A, two Tier B, five Tier C, two comparators; Supplementary Table S3); the Tier A $\cup$ Tier B set (five pairs) constitutes the primary literature-derived benchmark. The evaluation frame spans the three gynecological lineages (Ovarian, Endometrial, Cervical) with ≥ 5 mutant and ≥ 5 wild-type cell lines per entry, yielding 110 driver $\times$ paralog $\times$ lineage entries (8 positive; 102 unlabeled controls). Performance is reported under three aggregation frameworks defined before final reporting (see “Evaluation frameworks”). **d**, Component decomposition: signed DD (AUROC = 0.629) versus PCS (0.825), Δ Expression alone (0.547), and necessity (0.642). Inset: paired bootstrap distribution of PCS minus DD (10,000 resamples; solid line, mean $+0.195$; dashed lines, 95% CI -0.132 to $+0.545$; the difference is not significant). The pair-level bootstrap and negative control are in Supplementary Fig. S2.

Proteomic context: paralog protein co-variation

Whether paralog compensation acts at the transcript or protein level matters. We analyzed quantitative proteomics from seven independent CPTAC cohorts: BRCA ($n = 122$), COAD ($n = 97$), LUAD ($n = 110$), GBM ($n = 99$), PDAC ($n = 140$), UCEC ($n = 95$), and LUSC ($n = 108$), totaling 771 tumor samples with matched protein abundance [29–31].

Protein-level paralog co-variation was detectable across cohorts (Fig. 2a). EP300↔CREBBP showed the most consistent signal: significant in 5 of 7 cohorts after within-pair BH correction (BRCA: $r = 0.263$, $q = 0.005$; COAD: $r = 0.510$, $q < 10^{-3}$; LUAD: $r = 0.380$, $q < 10^{-3}$; GBM: $r = 0.501$, $q < 10^{-3}$; LUSC: $r = 0.372$, $q < 10^{-3}$; mean $r = 0.320$; Fig. 2b). Across all 26 pairs $\times$ 7 cohorts, 122 tests were evaluable with ≥ 10 paired protein measurements (of 182 possible), and 42 were significant at global BH $q < 0.05$ (per-cohort scatter plots in Supplementary Fig. S4). KRAS↔HRAS had the strongest single-cohort correlation ($r = 0.667$, $p < 10^{-4}$ in LUSC). Protein co-variation alone does not prove compensatory dependency; it could reflect shared transcriptional regulation, tumor purity, or copy number linkage. These results provide orthogonal contextual evidence of co-variation, not direct validation of conditional dependency.

Matched mRNA co-variation was also detectable. Across 179 pair $\times$ cohort mRNA tests covering the same 26 pairs in the seven cohorts, the median mRNA correlation was $r = 0.074$, versus $r = 0.140$ for the 122 pair $\times$ cohort tests with matched protein measurements; protein correlations were not systematically stronger (sign test on $|r|$, $p = 0.79$), and for EP300↔CREBBP the mRNA correlation exceeded the protein correlation in all seven cohorts (e.g., UCEC $r = 0.765$ vs. $r = 0.081$). Adjusting protein correlations for tumor purity where annotation exists (BRCA, LUAD, LUSC, UCEC; 69 tests) changed them negligibly (median $|\Delta r| = 0.014$; no test lost significance after adjustment), so the protein co-variation is not a purity artifact. As an SL-discrimination feature, however, the RNA layer remained weak: Δ Expression alone achieved AUROC = 0.547 for discriminating known paralog-SL pairs (Fig. 2c), below DD (0.629): a co-variation-versus-discrimination distinction that may contribute to the underperformance of transcript-only predictors. To test directly for compensatory upregulation, we compared ARID1B protein abundance in ARID1A-mutant ($n = 37$) versus wild-type ($n = 58$) endometrial tumors in the UCEC cohort. ARID1B abundance was higher in mutant tumors, but the difference was not statistically significant (log₂ difference 0.063, fold change ≈ 1.05 ; bootstrap 95% CI -0.018 to $+0.141$; rank-biserial $r = 0.213$; Wilcoxon $p = 0.082$, BH $q = 0.27$ across the ten evaluable UCEC mutation-conditioned tests; Fig. 2d). The direction of the effect, however,

279 is consistent with a model in which loss of one complex subunit stabilizes its paralog.

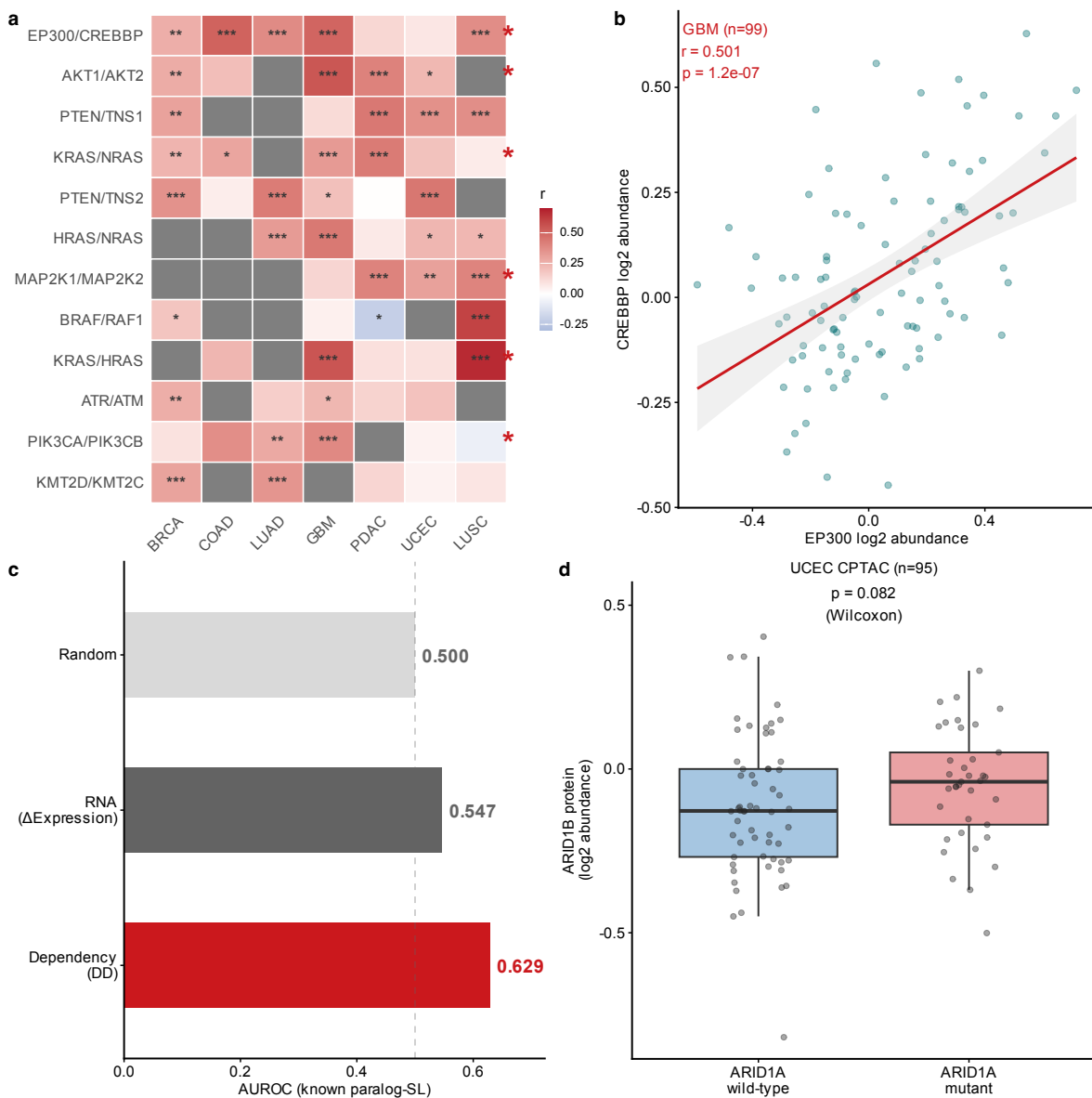


Fig. 2: CPTAC proteomic co-variation across seven cohorts. a, Cross-cancer paralog protein co-variation (Pearson's r) across seven independent CPTAC cohorts ($n = 771$ total). Significance: *** $q < 0.001$, ** $q < 0.01$, * $q < 0.05$ (BH-corrected within each pair). Blue cells with significance markers indicate significant negative correlations. Known SL pairs marked with *. b, Representative scatter plot: EP300 vs. CREBBP protein abundance in the GBM cohort ($n = 99$, $r = 0.501$, $p < 10^{-4}$). Gray band, 95% CI of linear regression. c, Dependency-based (signed DD, AUROC = 0.629) vs. RNA-level (Δ Expression, AUROC = 0.547) prediction performance for known paralog-SL pairs. The RNA signal is weak (dashed line, random expectation). d, Mutation-conditioned proteomics: ARID1B protein abundance in ARID1A-mutant ($n = 37$) vs. wild-type ($n = 58$) endometrial tumors (UCEC CPTAC cohort). ARID1B was higher in ARID1A-mutant tumors (fold change ≈ 1.05 ; Wilcoxon $p = 0.082$, not significant), a directionally concordant but statistically inconclusive signal.

280 DD signal across clinical contexts: MSI status and mutation type

281 Microsatellite instability (MSI) is a routine clinical biomarker in colorectal and endometrial
 282 cancers, and MSI-H tumors accumulate hypermutation-driven passenger events. We tested

whether this mutational background dilutes paralog compensation signals by classifying cell lines with the official DepMap microsatellite annotation (MSIsensor2 MSIscore; MSI-H = score > 20; Methods) and computing DD within each subgroup. On the ≥ 3 -per-group sensitivity frame (Fig. 3a), MSI-H and MSS subgroups showed comparable paralog-SL predictive signal in both cancer types: endometrial AUROC = 0.303 (MSI-H, $n = 17$) vs. 0.500 (MSS, $n = 11$), $\Delta = -0.197$; colorectal AUROC = 0.558 (MSI-H, $n = 14$) vs. 0.545 (MSS, $n = 45$), $\Delta = +0.013$. On the primary ≥ 5 frame the endometrial subgroups are not evaluable (fewer than two curated positive pairs) and the colorectal subgroups rest on two positives each (0.574 MSI-H vs. 0.298 MSS), so the MSI contrast is strictly exploratory. Subgroup sizes are modest and only 3–4 curated positive pairs were evaluable per subgroup; a stratified bootstrap test (10,000 resamples preserving within-subgroup label structure) found no significant MSI-H versus MSS difference in either cancer type (endometrial Δ AUROC = -0.197 , bootstrap $p = 0.534$; colorectal $\Delta = +0.013$, $p = 0.967$). A higher-powered formulation avoids subgrouping altogether: per-pair ordinary least-squares models of paralog gene effect on driver mutation $\times$ MSI-H status, fitted on all annotated lines of each lineage (colorectal $n = 63$, endometrial $n = 28$; Methods), tested directly whether DD itself varies with MSI background. Across 47 evaluable driver $\times$ paralog pairs in colorectal and 24 in endometrial cancer, no pair showed significant MSI modulation after Benjamini–Hochberg correction (minimum $q = 0.157$ and 0.235 , respectively); interaction coefficients were small (median $|\Delta$ DD| 0.062 in colorectal and 0.076 in endometrial) and all but four (colorectal) and one (endometrial) of the pair-level 95% confidence intervals spanned zero; the counts of nominally significant interactions did not exceed the 5% chance rate (4 and 1 pairs; binomial $p = 0.207$ and 0.708), and combined tests were null (Stouffer $p = 0.667$ and 0.276). For the lead candidate ARID1A $\rightarrow$ ARID1B, the MSI-H versus MSS difference in DD was negligible in both lineages (endometrial Δ DD = $+0.02$, $p = 0.96$; colorectal $+0.04$, $p = 0.79$), and a continuous formulation using the MSIsensor2 score gave concordant results (Methods). We therefore find no evidence that a hypermutated background either dilutes or amplifies paralog-SL detection by DD; these analyses remain hypothesis-generating.

Mutation type further modulated the signal. For the leading TSG pair, truncating mutations produced substantially stronger DD than missense variants (ARID1A $\rightarrow$ ARID1B in Ovarian: DD = 0.388 truncating vs. 0.020 missense, Δ DD = $+0.368$; Fig. 3b), and in Breast cancer truncating-only and missense-only DD performed similarly (AUROC = 0.460 vs. 0.437, both below 0.5; Supplementary Fig. S5). The pattern was not universal, however: across all tested

315 pairs, mean $|DD|$ was modestly larger for missense than for truncating subgroups (0.042 vs. 0.026;
316 paired t -test $p = 3 \times 10^{-4}$), driven mainly by oncogene pairs such as PIK3CA $\rightarrow$ PIK3CB in which
317 hotspot missense mutations define the mutant group (Supplementary Fig. S5b). These are point
318 estimates from small per-subgroup samples and should be treated as descriptive observations.

319 As an exploratory clinical correlation, we tested whether paralog expression tracks patient
320 outcome in TCGA. In the BRCA cohort ($n = 1,069$, 151 events), higher ARID1B expression
321 was associated with shortened overall survival (multivariable Cox HR = 1.31 per SD, 95% CI:
322 1.087–1.569, $p = 0.004$, adjusted for age and AJCC stage; Supplementary Fig. S10), though the
323 association reached FDR significance only after covariate adjustment and does not establish a
324 relationship between ARID1B expression and ARID1A-conditioned dependency. Four of the 32
325 paralog genes passed Benjamini–Hochberg correction (full results in Supplementary Table S8).
326 SMARCA2 showed no association (HR = 0.96), a reminder that expression-level prognosis and
327 dependency-based synthetic lethality are distinct phenotypes; these associations are hypothesis-
328 generating. Cohort-specific extensions in UCEC and OV are reported in the Supplementary
329 TCGA survival extension. Mutational co-occurrence analysis showed that paralog pairs co-occur
330 rather than being mutually exclusive, but the pan-cancer signal is partly confounded by tumor
331 mutational burden (TMB): after TMB adjustment only ARID1A/ARID1B (adjusted OR = 2.38,
332 $p = 0.002$) and EP300/CREBBP (adjusted OR = 2.14, $p = 0.011$) remained significant (per-
333 lineage analysis in Supplementary Fig. S6). These observations motivate analysing dependency
334 phenotypes directly but do not constitute evidence for synthetic lethality.

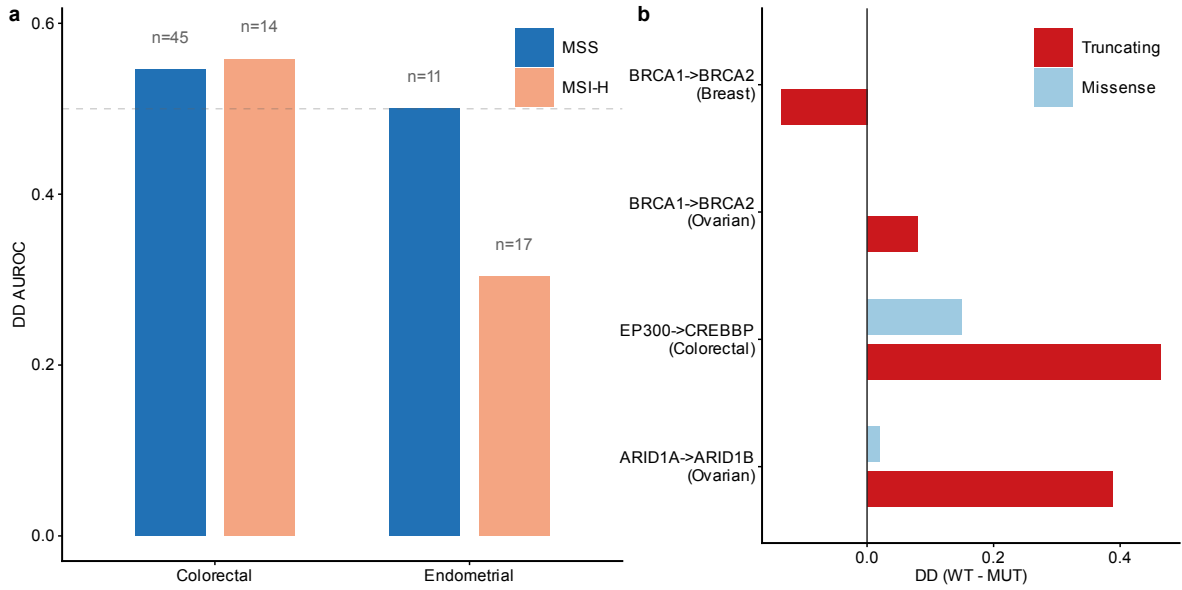


Fig. 3: Clinical stratification by MSI status and mutation type. a, DD AUROC for MSI-H vs. MSS subgroups in colorectal and endometrial cancers, using the official DepMap MSI-sensor2 annotation (MSI-score > 20 = MSI-H), computed on the ≥ 3 -per-group sensitivity frame (the primary ≥ 5 frame is not evaluable for endometrial and shows no colorectal difference; see text). Sample sizes shown above bars. MSI-H and MSS subgroups showed comparable DD signal in both cancer types (stratified bootstrap, not significant; a higher-powered full-cohort interaction regression likewise found no significant MSI modulation of DD after FDR correction; see text). b, DD (WT - MUT) for truncating vs. missense mutations in leading TSG paralog-SL pairs across four cancer types. Positive values indicate compensation; larger truncating bars indicate stronger compensation under truncating mutations. See Supplementary Fig. S5 for full results.

Dependency-window scoring and pharmacologic context prioritize ARID1A→ARID1B

To assess pharmacologic relevance, we analyzed the PRISM Repurposing dataset (1,482 compounds across 727 cell lines) [35]. At a discovery-stage threshold (BH $q < 0.25$, $\Delta\text{AUC} < 0$, enrichment > 0.1; Supplementary Fig. S7), 633 driver-genotype–drug sensitivity associations passed. By definition of the BH q -value, up to $\sim 25\%$ of associations passing this cutoff are expected to be false positives; the relaxed threshold is deliberate because this is a screen, not a confirmatory analysis; proliferation rate was not controlled, so these associations are quality-control context only. Encouragingly, known drug-target biology was recovered (Supplementary Fig. S7b): MEK inhibitors killed KRAS-mutant lines (AZD8330: $d = -0.62$, $\Delta\text{AUC} = -0.37$, $q = 0.0007$; Trametinib: $d = -0.57$, $\Delta\text{AUC} = -0.24$, $q = 0.0006$), mTOR and AKT inhibitors killed PTEN-mutant lines (Everolimus: $d = -0.61$, $\Delta\text{AUC} = -0.32$, $q = 0.0009$; Ipatasertib: $d = -0.99$, $\Delta\text{AUC} = -0.14$, $q = 0.0001$), and HDAC inhibitors killed EP300-mutant lines (Panobinostat in Ovarian cancer: $d = -1.45$, $\Delta\text{AUC} = -0.34$, $q = 0.010$). These associations recover expected driver-genotype drug biology and serve as an assay-validity control for the PRISM

layer; they do not demonstrate paralog-specific targeting.

To distinguish context-selective paralogs from pan-essential ones, we defined a dependency window score (DWS; an in vitro dependency-selectivity ratio, not a clinical dose-toxicity therapeutic index):

$$\text{DWS} = \frac{\max(\text{DD}, 0)}{\max(|\mu_{\text{Chronos}}|, f_{\text{pan-essential}}, 0.01)} \quad (2)$$

where $f_{\text{pan-essential}}$ is the fraction of cell lines with paralog Chronos score < -0.5 , and selectivity is the mutant-minus-wild-type difference in that fraction between driver-mutant and wild-type groups. The signed numerator credits only compensation-direction shifts; the $|\text{DD}|$ -numerator variant, which also credits reverse-direction shifts, is reported as a sensitivity analysis (Supplementary Table S10). Each paralog was classified into one of four selectivity tiers (Fig. 4). Two pairs fell into HIGH_SELECTIVITY: ARID1A→ARID1B (mean DWS = 2.82 across four cancer contexts; mean selectivity +0.28) and the established synthetic-lethal pair SMARCA4→SMARCA2 (4.87; +0.18), whose recovery acts as a positive control for the framework. This is what one would expect if ARID1A loss creates a specific dependency on ARID1B-containing BAF complexes [19]: within the dependency-window module ARID1A→ARID1B attained the highest mean selectivity and the highest mean positive DD (0.270) of all evaluated pairs (Supplementary Table S5). NF1→RASA2 attains a moderate signed DWS (1.82) that derives entirely from a single context (Endometrial), where the 0.01 denominator floor is active; its bootstrap rank interval is wide (95% CI 1–18) and selectivity is ≈ 0 , so NF1→RASA2 is not a selective candidate. BRCA1↔BRCA2 and PIK3R1→CRKL were classified as PAN_ESSENTIAL (pan-essential fraction 0.53–0.55 and 0.73), suggesting that targeting them directly would cause broad cellular toxicity. Five paralog pairs maintained DWS > 1.0 in ≥ 2 cancer contexts (Supplementary Fig. S8).

As an exploratory complement, we computed sequence- and structure-derived targetability descriptors for 13 paralog pairs (Supplementary Fig. S9; Methods). Five pairs (38%) showed perfect domain-architecture conservation, and EP300↔CREBBP was the most similar pair (0.976), consistent with shared histone acetyltransferase function. A composite score with heuristically fixed weights (max-normalized DWS 0.40, rescaled selectivity 0.30, targetability 0.30) ranked ARID1A→ARID1B first (0.823), followed by EP300→CREBBP (0.644) and NF1→RASA2 (0.554), making ARID1A→ARID1B the highest-priority established mutation-conditioned pair under the combined criteria (Supplementary Table S9). These descriptors are heuristic proxies,

not a formal druggability assessment, and were not calibrated against drug-development data
(Methods; Limitations).

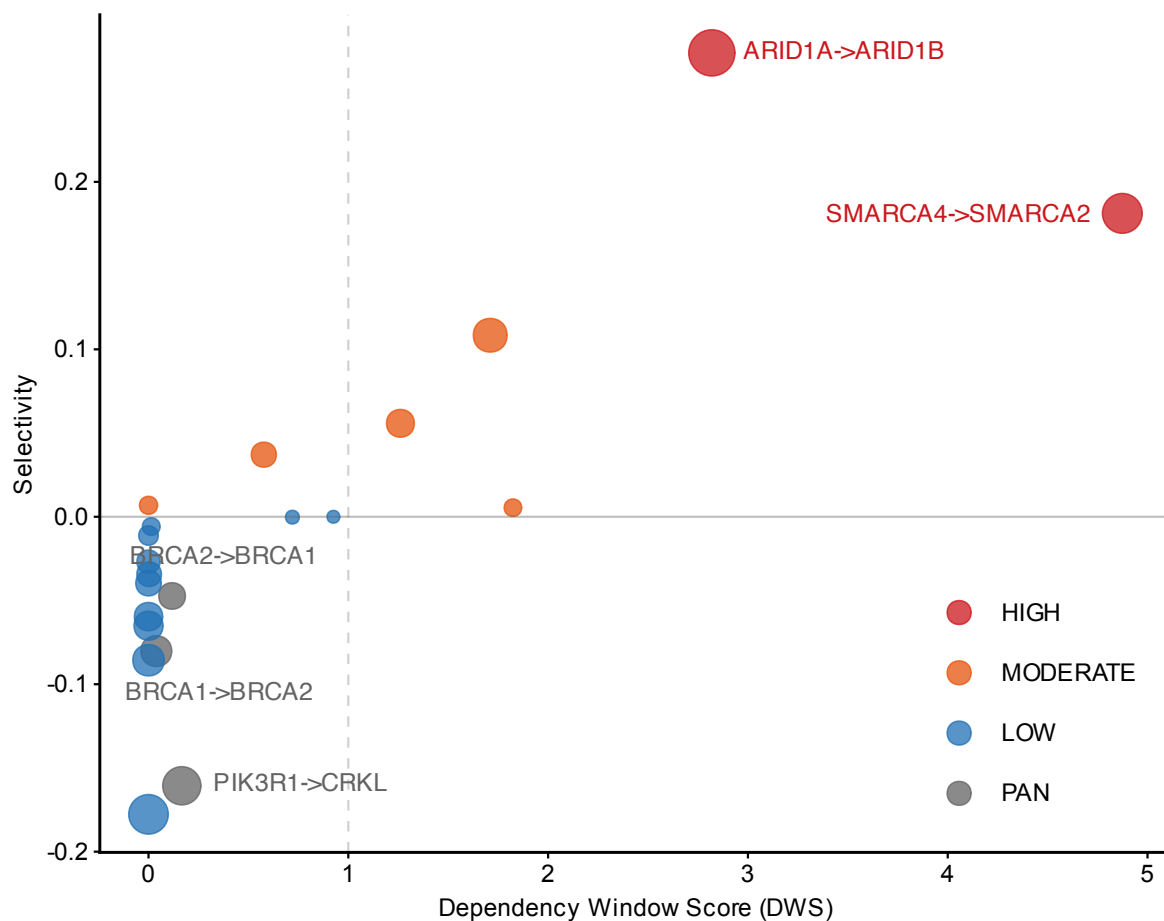


Fig. 4: Dependency-window classification. Bubble size, mean DWS; color, selectivity tier ($n = 21$ paralog pairs). HIGH_SELECTIVITY requires selectivity > 0.15 and DWS > 1.0 ; NF1→RASA2 has moderate signed DWS (1.82) resting on a floored denominator but near-zero selectivity (0.005), hence MODERATE. See Supplementary Fig. S8 for the full classification and Supplementary Fig. S9 for the exploratory structure-derived descriptors; PRISM pharmacologic context is shown in Supplementary Fig. S7.

Discussion

We set out to test whether a single, interpretable metric, the dependency shift between driver-mutant and wild-type cells, could capture the paralog-SL prioritization signal. DD provided a transparent univariate signal, whereas multi-feature performance estimates were unstable because only six positive pairs were available. On the 72-pair benchmark frame (three gynecological lineages), the interpretable composite score reached AUROC = 0.831 in head-to-head comparison, with no detectable difference from the best multi-feature classifier (SVM-RBF, 0.841; classifier range 0.240–0.841; paired-bootstrap Δ AUROC = -0.01 , 95% CI -0.21 to $+0.18$, too wide to establish equivalence), while DD alone reached 0.672 on the same mean-aggregated pair

frame (0.598 with max aggregation; “Evaluation frameworks”). The dependency shift carries a substantial but not dominant share of the signal; PCS and expression features contribute complementary information.

Protein and RNA co-variation. Our seven-cohort CPTAC survey found consistent paralog protein co-variation: EP300↔CREBBP was significant in five of seven cohorts, and KRAS↔HRAS reached $r = 0.667$ in LUSC. The protein signal is robust to tumor-purity adjustment (median $|\Delta r| = 0.014$ across 69 tests). Matched mRNA co-variation is also present, however: for EP300↔CREBBP the mRNA correlation exceeded the protein correlation in every cohort, and across the 122 matched pair×cohort tests, protein correlations were not systematically stronger than mRNA correlations (sign test on $|r|$, $p = 0.79$). A pure complex-stoichiometry model, in which loss of one subunit stabilizes its paralog strictly at the protein level, therefore cannot by itself account for the observed co-variation, which more likely reflects co-regulation at both layers; Venkatesh et al. attribute much paralog protein compensation to hemizygous copy-number loss rather than point mutation [33]. What separates the layers in our data is discriminative rather than correlative signal: RNA-level Δ Expression was a weak SL feature (AUROC = 0.547), whereas the dependency-layer statistic (DD) captures post-transcriptional essentiality biology directly. The mutation-conditioned analysis in UCEC was directionally concordant but statistically inconclusive: ARID1B protein abundance was higher in ARID1A-mutant tumors (fold change ≈ 1.05 , $p = 0.082$, BH $q = 0.27$), and the result needs replication in additional cohorts and pairs. Whether protein abundance itself could substitute for dependency data as a prioritization layer is untested, and PROTAC-based strategies that exploit protein abundance rather than transcriptional regulation remain a plausible but untested direction.

Directionality: TSG versus oncogene contexts. Signed-DD performance did not differ significantly between TSG-driven (mean AUROC = 0.649, $n = 9$) and oncogene-driven (0.834, $n = 3$) lineages on the ≥ 3 -per-group sensitivity frame (permutation $p = 0.127$); with only three oncogene lineages the comparison is underpowered, and the primary ≥ 5 frame retains a single oncogene-driven lineage, precluding estimation there. The more informative contrast is at pair level: TSG loss-of-function pairs show the expected compensation direction (positive DD; e.g., ARID1A→ARID1B, DD = 0.386 in Ovarian cancer), whereas the oncogene pair PIK3CA→PIK3CB shows a consistent reverse-direction shift (negative DD in all eight evaluable primary-frame lineages; e.g., DD = −0.258 in Breast cancer). Cells carrying an activated PIK3CA allele are less, not more, dependent on the paralogous catalytic subunit, plausibly

because the mutant isoform itself sustains downstream signalling. Signed DD makes this mechanistic distinction explicit; scoring by $|DD|$ would count such anti-dependencies as hits, which is a principal reason we report signed DD as the primary metric.

Clinical implications. A full-cohort interaction analysis found no evidence that MSI status modulates DD signal strength in the colorectal or endometrial lineages (Fig. 3a; no pair significant after FDR correction), although power remains limited for rare driver strata, whereas truncating mutations elicited stronger compensation than missense variants for several TSG pairs. These observations generate testable hypotheses for experimental work rather than a clinical enrichment strategy: truncating ARID1A mutations mark the context in which compensation is strongest, and baseline ARID1B abundance could be examined as a correlate of dependency in isogenic models; the data provide no basis for stratifying by MSI status, and this negative result warrants re-estimation in larger cohorts. Our DWS framework nominated ARID1A→ARID1B (DWS = 2.82) as the highest-selectivity established candidate (SMARCA4→SMARCA2 had the higher DWS, 4.87) while flagging BRCA1/2 paralogs as pan-essential. The DWS is based entirely on in vitro data and cannot predict organ-specific in vivo toxicity; natural extensions would incorporate tissue expression data from GTEx, mouse knockout phenotypes from IMPC, and germline constraint metrics (pLI/LOEUF) from gnomAD. The SMARCA4→SMARCA2 precedent illustrates both the feasibility and the demands of such a translational path: selective degradation of SMARCA2 was first achieved with VHL-based PROTAC degraders [20, 21], and the clinical-stage SMARCA2-selective degrader PRT3789 reduced SMARCA2 protein in patients’ peripheral blood mononuclear cells and produced RECIST-confirmed partial responses in biomarker-selected SMARCA4-mutated cancers [22]. That path required a dedicated bromodomain ligand, ternary-complex optimization, and extensive medicinal chemistry, none of which our computational targetability descriptors attempt to model [6].

Relationship to prior work. D’Antonio et al. [17] published the foundational analysis of paralog genetic interactions, and De Kegel et al. [50] systematically predicted robust paralog synthetic lethality across hundreds of cancer cell lines using interpretable features (protein interaction, evolutionary conservation); well-established pairs whose drivers fall outside our 40-gene panel, such as STAG2→STAG1 [60], reinforce the generality of paralog compensation but are not evaluable in our driver-conditioned framework. Baseline essentiality features carry much of the predictive signal in dependency-based models [56, 57], consistent with our finding that the necessity component alone reaches AUROC = 0.642. Combinatorial CRISPR platforms have

454 since mapped paralog synthetic lethals experimentally [32, 52–54, 61, 62], converging on 13 can-
 455 didate cross-study gold standards [32] whose replication across screens remains imperfect [51].
 456 Our work is complementary to these efforts in three ways. We benchmark DD against eight
 457 published deep learning methods in the CV3 setting, a comparison their study did not perform.
 458 Our seven-cohort CPTAC proteomic survey reveals protein-level co-variation that their analysis
 459 did not examine. Venkatesh et al. [33] independently demonstrated widespread paralog protein
 460 compensation in CPTAC tumors driven mainly by hemizygous copy-number loss, echoing the
 461 CYCLOPS paradigm of collateral dependency from passenger deletions [58, 59]; our analysis is
 462 complementary in that we condition on driver point mutations rather than copy-number loss
 463 and couple the proteomic signal to dependency-based prioritization. And we add stratification
 464 analyses (MSI, mutation type) plus a quantitative dependency-window framework for trans-
 465 lational prioritization. Together, the five orthogonal evidence layers go beyond a gene-family
 466 dependency atlas toward a prioritized candidate list for experimental follow-up. Two combina-
 467 torial screens published in 2025 extend this map at much larger scale: Harle et al. [64] tested
 468 472 nominated pairs in 27 cell lines and validated 117 synthetic-lethal interactions, and Flister
 469 et al. [65] screened the near-complete paralogome (36,648 pairs) in two models, identified 146
 470 lethal pairs, and trained predictive models on screen-derived features. Our contribution is or-
 471 thogonal to both: rather than expanding the digenic map or fitting multi-feature predictors, we
 472 show that a single transparent quantity computable from existing monogenic dependency atlases
 473 recapitulates digenic signal in both screens, while making explicit that the driver-conditioned,
 474 directional component of the proposition cannot be adjudicated at current screen scale.

475 Data independence. Both DD and the curated labels come from DepMap CRISPR data,
 476 which raises a circularity concern. We addressed this with a three-pronged strategy. First,
 477 the headline estimate is robust to direction-unsupported labels but attenuates when DepMap-
 478 era labels are removed: relabelling the pair whose direct evidence supports only the reciprocal
 479 direction (EP300→CREBBP) as non-positive left AUROC unchanged at 0.629, whereas remov-
 480 ing the two DepMap-derived pairs reduced it to 0.613 and restricting to the eight pairs with
 481 pre-DepMap experimental evidence gave 0.525. Because the two DepMap-derived labels were
 482 themselves curated from this dataset, their direction consistency is partially expected, and the at-
 483 tenuation after their removal quantifies how much of the headline signed-direction signal rests on
 484 dataset-internal curation rather than pre-existing experimental evidence. Both lineage-evaluable
 485 positive entries on the Tier A $\cup$ Tier B literature-derived benchmark (ARID1A→ARID1B in

two lineages) ranked above all unlabeled controls; the remaining benchmark pairs had too few qualifying driver-mutant lines for evaluation. Second, CPTAC proteomics, PRISM drug sensitivity, and TCGA survival provide independent data layers. Third, prospective validation against SL pairs catalogued after the DepMap 26Q1 release is planned.

The ARID1A→ARID1B nomination is evidence integration, not a significance claim. Under the within-driver multiple-testing framework (Benjamini–Hochberg correction across each driver’s HGNC paralogs), ARID1A→ARID1B does not pass the discovery threshold ($q = 0.39$ in Ovarian; $q = 0.88$ in Endometrial): ARID1A has >200 annotated paralogs, so the within-driver family is extremely large and power is limited. We therefore make no family-wise significance claim for this pair. Because ARID1A→ARID1B is itself a Tier B benchmark pair, its recovery serves as a second positive control for the framework (after SMARCA4→SMARCA2), not as a novel discovery. Its nomination rests instead on convergent, complementary criteria: the largest lineage-level effect size among tested pairs (DD = 0.386 in Ovarian), the leading dependency-window profile among selective candidates (DWS = 2.82; mean selectivity +0.28), and strong prior biological evidence [19]. The multivariate regression reported above (association of ARID1B dependency with ARID1A mutation after TP53 and CNV adjustment) answers a different question from the within-driver FDR test, conditional association rather than family-wise enrichment, and we do not use its low p-value to override the FDR result. ARID1A→ARID1B should thus be read as a high-priority validation candidate whose statistical evidence is incomplete, not as a statistically confirmed synthetic-lethal pair.

Beyond the benchmark: next-tier candidates. Because the two HIGH_SELECTIVITY pairs are both established synthetic lethals, the framework’s discovery value rests on its unlabeled candidates. The highest-ranked non-benchmark candidate is KMT2D→KMT2C (signed DWS = 1.26, MODERATE selectivity +0.056; composite score 0.525, fourth overall): KMT2D is one of the most frequently mutated epigenetic regulators across solid tumors, KMT2C and KMT2D are the catalytic paralogs of the COMPASS-like H3K4 methyltransferase complexes, and the pair shows high sequence-level similarity (0.829) with conserved domain architecture. Its dependency shift is modest ($|DD| = 0.064$), so we nominate KMT2D→KMT2C for combinatorial-CRISPR testing rather than claiming a validated dependency. This pair was also included in the unbiased digenic screen of Flister et al. [65] and was not called lethal in either of the two models screened (NCI-H1299, MDA-MB-231), consistent with the low genotype penetrance of digenic interactions and with the modest dependency shift we observe, but an external caveat

we report explicitly. RB1→RBL1 shows a reverse-direction shift, so its signed DWS is zero and it is not a selective candidate (its former |DD|-based DWS of 3.44 rested on a near-zero essentiality denominator with selectivity ≈ 0.007).

Evaluation frameworks. We report DD performance under three aggregation choices, all on the same benchmark universe: the twelve curated pairs plus unlabeled paralog controls, evaluated in the three gynecological lineages (Ovarian, Endometrial, Cervical) carried through the full pipeline (110 driver×paralog×lineage entries, 8 positive entries; 72 unique pairs, 6 positive pairs). (i) The lineage-level evaluation (AUROC = 0.629; pair-clustered bootstrap 95% CI 0.253–0.933; pair-level permutation $p = 0.122$; effective sample size ≈ 85 of 110 entries) preserves lineage-specific signed-DD values and is the primary framework. (ii) The per-pair evaluation with max aggregation (AUROC = 0.598) scores each pair by its largest signed DD across the three lineages; after the class-specific driver-mutation rules shrank the evaluable positive set, it does not significantly exceed the permutation null ($p = 0.223$), a power limitation we report explicitly rather than a framework we rely on. Bootstrap and permutation analyses use this framework. (iii) The per-pair evaluation with mean aggregation (AUROC = 0.672; permutation $p = 0.090$) averages signed DD across lineages and is the aggregation used for the per-pair classifier features (Methods). Frames (ii) and (iii) differ only in the aggregation function, and neither is directly comparable to (i).

Limitations.

1. Sample size: Several rare driver mutations appear in fewer than five DepMap cell lines and could not be analyzed in the primary framework; some lineages lacked sufficient samples for MSI and mutation-type stratification.
2. Evaluation-set heterogeneity: The evaluation set mixes functional analogs (e.g., BRCA1↔BRCA2, whose low k -mer Jaccard ≈ 0.23 reflects short 3-mers shared by chance between long proteins rather than true sequence homology) with true sequence paralogs. Stratifying by paralog type in future evaluations would improve clarity.
3. Mutation-agnostic gold standards: Conditioning on natural mutations of the 13 cross-platform in4mer gold-standard pairs [32], predominantly housekeeping genes whose alterations are mostly passenger events, did not separate them from unlabeled control pairs (10 of 13 pairs evaluable at the ≥ 3 -mutant threshold: AUROC = 0.50, permutation $p = 0.51$; only 3 pairs met the ≥ 5 threshold: exploratory AUROC = 0.68). The

three pairs with the largest mutant-line counts (ARFGEF1/ARFGEF2, HDAC1/HDAC2, SLC25A28/SLC25A37) each exceeded the control 90th percentile (Supplementary Fig. S2), suggesting DD recovers signal when mutations are abundant and functionally selected. DD is therefore scoped to driver-mutation-conditioned dependency; scored against direct combinatorial assays at adequate scale, mutation-agnostic |DD| does separate digenic hits from non-hits (Results), but those screens cannot test the driver-conditioned proposition itself (next item).

4. External digenic screens: The two combinatorial screens used as external tests contain few models carrying our driver mutations (three evaluable pairs per dataset; 9 and 6 mutant model-observations), so the genotype-stratified test is uninformative rather than confirmatory or contradictory. Both datasets were publicly released before we accessed them (2023–2024 data releases and preprint) and were analysed without modifying any frozen pipeline component; they therefore constitute independent external validation but not a prospectively registered hold-out.
5. Circular validation: DD and the curated labels both derive from DepMap data; the core AUROC measures internal consistency, not independent prediction. CPTAC, PRISM, and TCGA provide partially independent support but do not directly validate SL prediction.
6. Observational design: DD captures association, not causation. We controlled for lineage, co-mutations, and CNV through multivariate regression but cannot eliminate residual confounding. CRISPR double-knockout in isogenic models remains the necessary next step.
7. In vitro therapeutic window: DWS is based on cell line dependencies and cannot predict organ-specific toxicity. Integration with GTEx, IMPC, and gnomAD constraint data is needed.
8. Structural analysis: Limited to sequence-level features and ESMFold-predicted single-chain structures truncated to 400 residues; the reported “structural similarity” is a feature-space heuristic ($1 - \overline{|\Delta\text{feature}|}$), not a three-dimensional alignment metric (no TM-score or RMSD), and the descriptors are not calibrated against drug-development data. AlphaFold-Multimer complex predictions and experimental binding data are needed for structure-guided drug design.
9. Positive-unlabeled evaluation: Unlabeled controls are HGNC paralog pairs without cu-

rated SL evidence, not experimentally verified non-SL pairs. AUROC estimates therefore measure separation from unlabeled pairs and are optimistic relative to a design with validated negatives.

10. Evaluation units: AUROC is computed at different units across analyses: driver $\times$ paralog $\times$ lineage entries in the primary framework versus aggregated pairs in the per-pair framework; the two are not directly comparable, and pair-level aggregation discards lineage structure.
11. Benchmark scope: The evaluation universe comprises three gynecological lineages (Ovarian, Endometrial, Cervical); per-lineage AUROCs outside this universe (Fig. 1a) rest on few positives each and are descriptive rather than independent benchmarks.
12. Effect direction: The primary benchmark ranks pairs by signed DD, so reverse-direction shifts (weaker paralog dependency in driver-mutant lines) count against a pair; the direction-agnostic $|DD|$ variant, which can admit such pairs, is reported as a sensitivity analysis.
13. Driver-mutation definitions: TSG and oncogene mutation calls follow different rules (LikelyLoF vs. Hotspot), so “mutation” is not a uniform intervention across gene classes; amplification-driven oncogenes (e.g., CCNE1) contribute no mutant lines under these rules and are only analyzed through copy-number channels.
14. CPTAC coverage: Not all paralogs were quantified in all cohorts (41–45 of 58 per cohort); mutation-conditioned analysis was performed only for ARID1A $\rightarrow$ ARID1B in UCEC.

Future directions. (1) Combinatorial-CRISPR validation of the nominated unlabeled candidates, starting with KMT2D $\rightarrow$ KMT2C in KMT2D-mutant versus wild-type isogenic lines, with viability and colony formation as primary readouts; for ARID1A $\rightarrow$ ARID1B, digenic knockout epistasis in ARID1A-mutant lines (e.g., TOV-21G) would extend the single-gene evidence of Helming et al. [19] to a dual-perturbation design. This study is purely computational; experimental validation through collaboration with functional genomics laboratories is the essential next step. (2) External hold-out validation against SL pairs catalogued in SynLethDB or other databases after the DepMap 26Q1 release. (3) Extending mutation-conditioned proteomic analysis from ARID1A $\rightarrow$ ARID1B in UCEC to additional paralog pairs and CPTAC cohorts. (4) Experimental assessment of ARID1B degradation strategies, informed by the SMARCA2-degrader precedent (Discussion), once suitable ARID1B ligands or degrader handles are identified. (5) Building lineage-specific paralog-SL atlases with MSI-aware annotation to support translational research.

Conclusions

Signed DD provides a direction-aware and experimentally interpretable statistic for prioritizing mutation-conditioned paralog dependencies, evaluated here across 23 solid tumor types. Its current performance estimates remain uncertain because the literature-derived positive set is small, heterogeneous, and partly dependent on DepMap-derived evidence. The framework recovered established mutation-conditioned dependencies, including ARID1A→ARID1B, the highest-selectivity established candidate, and prioritized KMT2D→KMT2C as the highest-ranked non-benchmark candidate for further testing, but neither DD nor the dependency-window score establishes causal synthetic lethality or an in vivo therapeutic window. Independent combinatorial perturbation in isogenic models is the essential next step.

We provide a complete open-source toolkit, the `paralogSL` R package and a reproducible Python pipeline, which together require only publicly available DepMap data and no specialized computational infrastructure.

Methods

Data sources and preprocessing

DepMap 26Q1. CRISPRGeneEffect.csv (420 MB; Chronos-normalized gene dependency scores (hereafter “gene effect scores”) for 1,208 cell lines × 18,531 genes), OmicsSomaticMutations.csv (554 MB), OmicsExpressionProteinCodingGenesTPMLogp1.csv (483 MB), and Model.csv (681 KB) were downloaded from the DepMap Portal (release 26Q1) [36, 40]. Driver-mutation status was defined by gene-class-specific rules on the default omics profile per model (`IsDefaultEntryForModel`) [41]: for tumor suppressors (TSG), a line is mutant if and only if it carries at least one DepMap-curated likely loss-of-function variant (`LikelyLoF`, which includes dominant-negative TP53 missense); for oncogenes, if and only if it carries at least one recurrent oncogenic hotspot variant (`Hotspot`); genes were classified as TSG or oncogene by the DepMap 26Q1 `OncogeneHighImpact/TumorSuppres` majority vote (one documented override: GATA3 as TSG, resolving a 20/20 majority-vote tie given its frameshift-dominated lesion spectrum; GATA3 drives none of the curated benchmark pairs). Amplification-driven oncogenes without qualifying hotspot mutations (e.g., CCNE1) contribute no mutant lines under this rule and are analyzed through copy-number channels instead. The complete per-gene variant classification table (old non-silent definition vs. class-specific rule, cell line counts, and variant-type distribution) is provided as Supplementary Table S6. In the

mutation-type analysis, multiple variants in the same gene per line are collapsed to the most severe consequence. Cell lines without expression or dependency data were excluded, yielding 1,208 cell lines. The exact files analysed (release filenames, sizes, SHA256 checksums, and per-file download dates) are tabulated in the repository data manifest (`data/README.md`), so the identical inputs can be retrieved and verified.

HGNC gene families. The HGNC complete set was downloaded from the HGNC repository [37]. Protein-coding genes with “Approved” status and non-empty `gene_group_id` were retained (11,050 genes). All pairwise gene group combinations yielded 66,595 paralog pairs.

CPTAC proteomics. Seven CPTAC cohorts were retrieved via the cBioPortal API [29, 30]: BRCA ($n = 122$), COAD ($n = 97$), LUAD ($n = 110$), GBM ($n = 99$), PDAC ($n = 140$), UCEC ($n = 95$), and LUSC ($n = 108$); these seven cohorts constitute the main proteomic analysis (771 tumors in total). Ovarian HGSC global proteome data (NCI Proteomic Data Commons study PDC000360, $n = 232$) [31] were downloaded for reference but were not used in the main analysis. Pearson correlations were computed on $\log_2$ -transformed abundance values (≥ 10 paired samples per test).

PRISM drug sensitivity. The PRISM Repurposing secondary screen dataset (1,482 compounds $\times$ 727 cell lines) was downloaded from the DepMap Portal [35]. $\Delta\text{AUC} = \text{mean}(\log_2\text{AUC}_{\text{MUT}}) - \text{mean}(\log_2\text{AUC}_{\text{WT}})$. Selective drugs: BH $q < 0.25$, $\Delta\text{AUC} < 0$, enrichment > 0.1 .

TCGA survival data. Overall-survival annotations and tumor mRNA expression for the TCGA Pan-Cancer Atlas breast cancer cohort were retrieved via the cBioPortal API (study `brca_tcga_pan_can_atlas_2018`) [29, 34]. Patients with both overall-survival and expression data ($n = 1,069$; 151 deaths) were analyzed. Cox proportional-hazards models (Wald 95% confidence intervals) were fitted on continuous $\log_2$ -transformed, z-scored expression, estimating hazard ratios per +1 SD for each of the 32 paralog genes from the 16 priority pairs; multivariable models adjusted for age and AJCC stage (ordinal I–IV). The proportional-hazards assumption was assessed with a Schoenfeld-type residual–time correlation test (Supplementary Table S8). The identical pipeline was extended to the UCEC ($n = 525$, 87 events) and OV ($n = 299$, 180 events) PanCan Atlas cohorts; AJCC stage annotation is defined but entirely unpopulated for these two studies in cBioPortal (verified at patient and sample level), so age-only adjusted models are the adjusted estimates of record for them (Supplementary Information, TCGA survival extension).

Mutational co-occurrence. For each priority driver–paralog pair, a 2×2 contingency table

of driver-rule mutation status (TSG: **LikelyLoF**; oncogene: **Hotspot**) was built across the 1,208 dependency-profiled DepMap 26Q1 cell lines. Co-occurrence odds ratios and two-sided p -values were computed with Fisher’s exact test; Wald 95% confidence intervals on $\log(\text{OR})$ used a 0.5 Haldane correction when any cell was zero.

Delta Dependency computation

DD was computed as defined in Equation 1, using Welch’s t -test (unequal variances); effect sizes are reported as Cohen’s d (pooled SD) and the small-sample-corrected Hedges’ g . Unless stated otherwise, DD is scored as a signed quantity (positive = compensation); analyses scored by $|\text{DD}|$, which additionally credits reverse-direction shifts, are reported as sensitivity analyses. The primary analysis required a minimum of five mutant and five wild-type cell lines per driver $\times$ lineage stratum; the more permissive $\geq 3/\geq 3$ threshold was re-run as a sensitivity analysis (Results). Benjamini-Hochberg correction [42] ($q < 0.20$) was applied within each driver gene: for driver D , all (D, P_i, c_j) combinations were corrected jointly. This is a discovery-stage pipeline; global FDR correction across all 66,595 pairs $\times$ 40 drivers $\times$ 23 cancer types was not performed because the goal was to maximize sensitivity for candidate identification. Candidates should be treated as hypothesis-generating. At a stricter $q < 0.10$, PIK3CA $\rightarrow$ PIK3CB remained significant ($q = 0.086$ in Endometrial), whereas ARID1A $\rightarrow$ ARID1B did not survive within-driver correction ($q = 0.50$), not because of weak signal ($|\text{DD}| = 0.30$) but because ARID1A has >200 HGNC paralogs.

Benchmark comparison

Curated pairs and unlabeled paralog controls. Twelve paralog-SL pairs were curated from published studies (Supplementary Table S3), which assigns each pair an evidence tier after verification of the primary citations. Tier A (three pairs: AKT1 $\rightarrow$ AKT2 [45], CDK4 $\rightarrow$ CDK6 and MAP2K1 $\rightarrow$ MAP2K2 [52]) comprises pairs with direct genetic synthetic-lethal evidence from dual-gene perturbation (combinatorial or digenic CRISPR knockout). Tier B (two pairs: SMARCA4 $\rightarrow$ SMARCA2 [18], ARID1A $\rightarrow$ ARID1B [19]) comprises pairs for which the paralog is a demonstrated selective dependency in driver-mutant cells (natural-genotype conditional dependency from single-gene perturbation with functional validation). The Tier A $\cup$ Tier B set (five pairs) constitutes the primary literature-derived benchmark; the two tiers differ in evidence independence: Tier A derives from dual-gene perturbation, whereas Tier B is itself natural-genotype

conditional dependency, the same data type DD measures, so Tier B concordance is partially expected by construction. Tier C (five pairs) comprises indirect evidence only and is excluded from the primary benchmark: EP300→CREBBP, established experimentally in the reciprocal direction only (CREBBP→EP300 [23, 48]); PIK3CA→PIK3CB, whose primary evidence supports PTEN→PIK3CB [25]; CCNE1→CCNE2, supported by developmental redundancy in mouse double knockouts [46]; and FBXW7→FBXW2 and PPP2R1A→PPP2R1B, derived from DepMap analyses. Two mechanistic comparators (BRCA1↔BRCA2 [4] and STK11→SIK1, an LKB1–SIK axis reference [47]) are not sequence paralogs and serve as specificity references only. Of the twelve pairs, 10 are true sequence paralogs (shared gene family, conserved domains), 1 is a partial homolog (STK11→SIK1), and 1 is a functional analog (BRCA1↔BRCA2). Unlabeled controls comprised all remaining HGNC-defined paralog pairs involving the same 40 driver genes that did not appear in the positive set; these controls are unlabeled rather than experimentally confirmed non-SL pairs, so the evaluation is a positive–unlabeled setting and AUROC quantifies discrimination against unlabeled controls, not against verified negatives. The primary ranking score is signed DD (Equation 1); the direction-agnostic |DD| variant is reported only as a sensitivity analysis because it also promotes reverse-direction effects (dependency decreasing after driver mutation, e.g., oncogene-addiction displacement for PIK3CA→PIK3CB). We audited every evaluation frame for directionality (`output/direction_audit.json`): on the Tier A ∪ Tier B benchmark set, signed and absolute scoring coincide (AUROC = 1.000), whereas on the full curated set 3 of 8 positive entries have DD < 0 (signed-DD AUROC = 0.629 vs. |DD| sensitivity 0.676); full-set sensitivity values should therefore be read as direction-agnostic discrimination, and directional claims rest on signed DD. The evaluation is highly imbalanced (12 curated positives vs. ~194 unlabeled controls per lineage). AUROC is insensitive to class prevalence, but precision–recall performance is not; we therefore report AUPRC (average precision) alongside AUROC, with bootstrap confidence intervals for both. Directional pairs (e.g., BRCA1→BRCA2 and BRCA2→BRCA1) are scored separately when both driver genes are in the 40-gene panel.

Contextual comparison with published methods.

DD was compared against eight published SL prediction methods evaluated in the CV3 setting by Feng et al. [14]. Published AUROCs are from the CV3 evaluation reported in Supplementary Data 1 of that study (NSMRand negative sampling, 1:1 positive:negative ratio, complete dataset); DD was evaluated on the paralog-SL prediction task using the 12 curated paralog-SL pairs as the positive set. AUROC was computed with the rank-based Mann–Whitney

statistic (identical to the scikit-learn implementation [43]); AUPRC is reported as average precision. Bootstrap 95% CIs used 10,000 resamples of evaluation entries (or pairs, for per-pair frames) with replacement; paired bootstrap resamples were used for head-to-head comparison of signed DD against component scores on identical resamples. For the DD + identity-proxy subset (3 pairs), bootstrap CI was computed separately to account for the reduced sample size. Negative controls: 100 permutations of shuffled known-SL labels.

External mutation-agnostic gold-standard check. The 13 cross-platform paralog gold standards from the in4mer digenic-knockout meta-analysis [32] were evaluated separately from the driver-conditioned benchmark because they are defined by double-knockout phenotype rather than by natural driver alterations. For each pair, both orientations (either member as the conditioning gene) were scored by the maximum |DD| across lineages, with mutation status defined by the pipeline’s gene-class rules (genes outside the 40-driver panel retain the permissive non-silent filter). Controls were 400 unlabeled paralog pairs sampled (seed 42) from the same Ensembl/HGNC pair universe, excluding the known-SL set and the in4mer pairs, and were scored identically. AUROC used the Mann–Whitney statistic with 1,000 bootstrap resamples (95% CI) and 10,000 label permutations (seed 42).

External digenic screens. Two published combinatorial CRISPR datasets were analysed as external tests with every pipeline component frozen before data access (code, features, mutation rules, and candidate list; engine fidelity against the frozen in4mer scoring: maximum absolute score difference 2.8×10^{-16} across 413 pairs). From Harle et al. [64] we used the long-format genetic-interaction table (their Table S4; 472 pairs $\times$ 27 lines) with their reported hit criterion (mean normalized genetic-interaction score < -0.5 and FDR < 0.01 , with neither single target depleted; a pair is a hit if hit in ≥ 1 line; 117/472 pairs). From Flister et al. [65] we used the per-model lethal calls (their Table S3; 146 pairs lethal in NCI-H1299 and/or MDA-MB-231) and, for genotype stratification, the 49-model interaction meta-analysis (their Table S6; hit if $\text{dzL2FC} \leq -2$, reproducing the study’s own SL counts for 98.3% of pairs). Where preprint text and tables differed on hit totals, table values were used. Each pair was scored by $\max|\text{DD}|$ across both orientations and all lineages under the ≥ 3 -mutant/ ≥ 3 -wild-type frame. Layer 1 (mutation-agnostic) compared scores between screen hits and non-hits by Mann–Whitney AUROC with 10,000 label permutations (seed 42). Layer 2 (genotype-stratified) projected the pipeline’s class-specific driver-mutation rules onto screen models via DepMap identifiers (26/27 Harle lines; 49/49 Flister models) and compared digenic hit rates (one-sided Fisher exact) and

interaction scores (one-sided Mann–Whitney) between driver-mutant and wild-type models, per pair and pooled, with Benjamini–Hochberg correction within each dataset. Analysis code: `external_holdout.py`; full outputs: `output/external_holdout.json` and companion CSV files.

Sequence identity computation

K -mer-based Jaccard similarity ($k = 3$) was used as a computationally efficient proxy for global protein sequence identity. This metric was validated against Needleman-Wunsch global alignment identity in a subset of 50 protein pairs (13 known paralog pairs and 37 non-paralog controls; Pearson $r = 0.88$, $p < 10^{-16}$; Supplementary Fig. S9). Within-group correlations were $r = 0.91$ (paralogs, $n = 13$) and $r = 0.41$ (non-paralogs, $n = 37$), confirming that the high overall r is not solely driven by between-group variance. The $\geq 30\%$ threshold should therefore be interpreted as an approximate filter for enriching high-identity pairs rather than a precise sequence identity cutoff.

Potential confounders and causal interpretation

DD quantifies an observational association between driver mutation status and paralog dependency, not a causal effect. Potential confounders include: (1) lineage differences between mutant and wild-type cell lines within a cancer type; (2) co-occurring mutations that may independently affect paralog dependency; (3) copy number alterations at paralog loci. We mitigated these through: within-lineage analysis (DD computed separately per cancer type), Welch’s t -test (robust to unequal variances), and CNV independence verification (Supplementary Fig. S3, all $R^2 < 0.10$). Benjamini-Hochberg correction was applied within each driver gene to control false discovery rate. Nevertheless, experimental validation (e.g., CRISPR double-knockout in isogenic cell lines) remains essential to establish causality.

CPTAC proteomic co-variation

Protein abundance data for seven CPTAC cohorts (BRCA, COAD, LUAD, GBM, PDAC, UCEC, LUSC; $n = 771$ total) were retrieved via the cBioPortal API [29, 30]. For each paralog pair, Pearson correlation of protein $\log_2$ -abundance was computed across all samples within each cohort. Significance was assessed using BH correction within each pair across 7 cohorts. For the RNA-layer comparison, continuous mRNA profiles (`rna_seq_v2_mrna`, log-scale values

used as provided) were retrieved for the same 26 pairs in all seven cohorts (179 pair×cohort tests with ≥ 10 samples), and protein–mRNA comparisons used the 122 pair×cohort combinations with both measurements. Tumor-purity robustness was assessed by partial Pearson correlation controlling for the cohort’s purity annotation (ESTIMATE tumor purity in BRCA, LUAD, and LUSC; CIBERSORT absolute score in UCEC; 69 tests with ≥ 15 matched samples). Mutation-conditioned proteomic analysis was performed for the ARID1A→ARID1B pair in the UCEC cohort: ARID1A mutation status was retrieved from the cBioPortal mutations API (`ucec_cptac_2020_mutations`), and ARID1B protein abundance was compared between ARID1A-mutant ($n = 37$) and wild-type ($n = 58$) samples using the Wilcoxon rank-sum test. Extension to additional pairs and cohorts is a priority future direction.

Pan-cancer lineage definitions

Twenty-seven solid tumor types were defined by matching OncotreePrimaryDisease annotations to curated keyword lists; the 23 types with ≥ 6 cell lines with available data were analyzed (Supplementary Table S1). Lineage short labels follow DepMap/Oncotree conventions and mix organ-based and histology-based names as required by each cancer type’s classification (e.g., NSCLC and SCLC are distinct histological subtypes of lung cancer); full names are given in Supplementary Table S1. Forty pan-cancer driver genes were curated from TCGA PanCanAtlas driver analyses [26] and the COSMIC Cancer Gene Census [49] (Supplementary Table S4); per-gene mutant-line counts under the final class-specific rules are reported in Supplementary Table S6 (CCNE1 contributes zero mutant lines under the hotspot rule; MAPK1, audited for the external in4mer analysis rather than as a panel gene, likewise yields zero).

MSI stratification

Cell lines were classified with the official DepMap 26Q1 microsatellite annotation (`OmicsGlobalSignatures.csv`). The DepMap consortium computes an MSI_{sensor2} MSI_{score} from each model’s sequencing data; following the consortium convention, lines with MSI_{score} > 20 were labelled MSI-H and all others MSS. Default-entry profiles (`IsDefaultEntryForModel`) were used; all 87 evaluable colorectal and endometrial lines had official annotations. Because the annotation is measured rather than mutation-inferred, POLE-mutant lines receive their observed status (15 of 22 POLE-mutant lines in these two lineages are officially MSI-H) and are not presumed hypermutated; agreement between this annotation and a damaging-mutation proxy (MLH1/MSH2/MSH6/PMS2/POLE)

was 89% (per-line audit at `output/msi_classification_audit.csv`). DD was computed separately within each subgroup for colorectal and endometrial cancers. For the higher-powered interaction analysis, per-pair ordinary least-squares models were fitted on all lineage lines with dependency data and official MSI annotation (colorectal $n = 63$, endometrial $n = 28$): $G_P \sim D_{\text{mutation}} \times \text{MSI-H}$, with driver mutation status from the gene-class rules described above; the interaction coefficient equals the MSI-H minus MSS difference in DD (ΔDD). A pair was evaluable with ≥ 3 mutant lines and ≥ 2 mutants in each MSI class (47 pairs in colorectal, 24 in endometrial). A continuous variant replaced the MSI-H indicator with $\log(1 + \text{MSIsensor2 score})$. Interaction p -values were combined per lineage with Stouffer’s method (signed by ΔDD), the excess of nominally significant interactions was assessed with a one-sided binomial test against the 5% null rate, and Benjamini–Hochberg q -values were computed within each lineage (`msi_interaction_test.py`; per-pair results at `output/msi_interaction_results.csv`).

Mutation type classification

Driver mutations classified as truncating (frameshift, nonsense, splice-acceptor, splice-donor, start-loss, stop-loss) or missense (`missense_variant`, `protein_altering_variant`, excluding truncating) using VEP VariantInfo annotations. Per-cell-line classification used the most severe mutation type.

Therapeutic window analysis

DWS defined as in Equation 2 with the signed numerator $\max(\text{DD}, 0)$, which credits only compensation-direction shifts; the $|\text{DD}|$ -numerator variant is reported as a sensitivity analysis (Supplementary Table S10). The denominator uses $\max(|\mu_{\text{Chronos}}|, f_{\text{pan-essential}}, 0.01)$ to capture the most conservative estimate of baseline paralog essentiality: $|\mu_{\text{Chronos}}|$ reflects mean dependency magnitude (dimensionless, normalized by Chronos), $f_{\text{pan-essential}}$ reflects the fraction of lines where the paralog is essential (dimensionless, range 0–1), and the 0.01 floor prevents division by near-zero values for non-essential paralogs. All three quantities are dimensionless ($f_{\text{pan-essential}}$ is bounded in 0–1; $|\mu_{\text{Chronos}}|$ is typically of order 0–1); the $\max$ operator selects whichever indicates the greatest baseline toxicity risk. Paralogs were classified with operational stratification thresholds: PAN_ESSENTIAL ($f_{\text{pan-essential}} > 0.5$), HIGH_SELECTIVITY (selectivity > 0.15 and DWS > 1.0), MODERATE, or LOW_SELECTIVITY, which are not validated clinical cutoffs (threshold sensitivity in Supplementary Table S10). Analysis across 5 cancer con-

856 texts (Ovarian, Endometrial, Breast, Colorectal, PanCancer). The module covers 24 candidate
857 pairs (13 curated-direction pairs, including the two comparators and the BRCA2→BRCA1 re-
858 verse orientation, plus 11 pipeline-nominated pairs); 21 had at least one context passing the ≥ 5 -
859 mutant frame and are reported, while AKT1→AKT2, CCNE1→CCNE2, and CDK4→CDK6
860 did not meet the threshold in any context. Robustness: pair rankings were re-derived under
861 alternative denominators ($|\mu_{\text{Chronos}}|$ only; $f_{\text{pan-essential}}$ only; the mean of the two), alternative
862 floors (0.001, 0.05), and alternative selectivity thresholds (0.10, 0.20), and bootstrap 95% CIs
863 for mean DWS and selectivity were estimated from 1,000 stratified resamples of cell lines (seed
864 42; Supplementary Tables S5 and S10; `dws_robustness.py`).

865 Exploratory sequence- and structure-derived targetability descriptors

866 Sequence-based features (length, molecular weight, pI, GRAVY hydrophobicity, disorder propen-
867 sity, lysine/cysteine density) were computed from UniProt canonical sequences, and domain ar-
868 chitectures from UniProt annotations. Structures were predicted with ESMFold v1 (`facebook/esmfold_v1`)
869 for the N-terminal 400 residues of 15 proteins (mean pLDDT = 0.724); for large proteins
870 this truncation covers only a fraction of the chain (e.g., $\sim 17\%$ for ARID1A/ARID1B), so
871 ARID1A/ARID1B were assessed with sequence-level features only. We emphasize the ex-
872 ploratory status of these descriptors. (i) The reported “structural similarity” is a feature-space
873 heuristic, $\max(0, 1 - \overline{|\Delta_{\text{feature}}|})$ over the sequence features above; it is not a three-dimensional
874 alignment metric (no TM-score or RMSD is used in the score). (ii) The “pocket count” is a cus-
875 tom sliding-window heuristic (windows with pLDDT > 0.7 and hydrophobic fraction 0.2–0.7,
876 aggregated per 12 windows), not a validated pocket-detection method. (iii) The PROTAC-
877 suitability score (lysine density $\times 2.0$ + disorder propensity – cysteine density $\times 0.5$) captures
878 only two of the many determinants of degrader efficacy; ligandability, E3-ligase compatibility,
879 ternary-complex geometry, and cell permeability are not modelled. A per-pair targetability score
880 combined structural similarity (0.25), domain-architecture conservation (0.30), the druggability
881 proxy (0.25), and the PROTAC score (0.20). The composite prioritization score combined max-
882 normalized DWS (0.40), selectivity rescaled to $[0, 1]$ as $(s + 1)/2$ (0.30), and targetability (0.30);
883 all weights were fixed heuristically before final reporting and are unchanged in the archived
884 release (commit 66fc633; `alphafold_analysis.py`), and none of the descriptors was calibrated
885 against external drug-development data.

Statistical analyses

AUROC via scikit-learn [43]. Bootstrap CI: 2.5th and 97.5th percentiles from 1,000 iterations. For the primary lineage-level frame, a pair-clustered bootstrap (10,000 iterations, resampling whole pairs together with all their lineage entries) and a pair-level label permutation test (10,000 permutations) account for within-pair correlation (one-way random-effects ANOVA ICC = 0.554 for signed DD within pairs; effective sample size 85 of 110 entries). A driver-block bootstrap (10,000 iterations, resampling driver genes together with all their pairs and lineage entries) further accounts for non-independence among pairs sharing a driver gene. Negative controls: 10,000 label permutations. Analyses in Python 3 with numpy [44], scipy [63], pandas, statsmodels, and scikit-learn [43]; exact dependency versions are pinned in the repository Dockerfile; the complete audit suite passes under the current managed runtime (Python 3.12, scikit-learn 1.9.0, numpy 2.4.4, scipy 1.18.0, pandas 3.0.2, statsmodels 0.14.6). The companion R package **paralogSL** provides DD computation, PCS analysis, benchmark comparison, and CPTAC correlation visualization under the MIT license.

Power analysis and test set stratification

Curated pairs were stratified into evidence tiers (Supplementary Table S3): Tier A (3 pairs; direct dual-perturbation genetic synthetic-lethal evidence), Tier B (2 pairs; natural-genotype conditional dependency with functional validation), Tier C (5 pairs; indirect, reciprocal-direction-only, or DepMap-derived evidence), and 2 mechanistic comparators (BRCA1 $\leftrightarrow$ BRCA2, STK11 $\rightarrow$ SIK1). The primary endpoint is the AUROC and AUPRC of signed DD on the Tier A $\cup$ Tier B literature-derived benchmark at the driver $\times$ paralog $\times$ lineage level, the natural unit at which DD is computed, with the pair treated as the grouping unit in sensitivity analyses to avoid cross-lineage information leakage. The full 12-pair curated set, tier-restricted, direction-strict, and per-pair aggregated analyses are reported as secondary and sensitivity analyses. With at most 2–5 evaluable positive pairs, subset-level AUROC has limited power: at $\alpha = 0.05$, an estimated ≥ 25 validated positive pairs would be needed to achieve 80% power for detecting AUROC = 0.85 against a null of 0.5. At the observed primary AUROC of 0.629, power with the current 8 positive and 102 control entries is only 0.34 (0.28 for the 6-positive, 66-control pair frame), and 80% power at this effect size would require ≥ 45 positive pairs (binormal model, Hanley–McNeil variance; **power_analysis.py**); because these calculations assume independent entries and verified true negatives, they serve as rough planning values under the positive–unlabeled design rather

than exact requirements. To compensate for this limitation, we complement binary classification metrics with effect-size ranking (DD, DWS, selectivity) and multivariate regression significance (p -values for individual pairs).

Supplementary information

Supplementary Figures S1–S10 and Supplementary Tables S1–S12 are available as a separate PDF file (Additional file 1); machine-readable TSV mirrors of the data tables are provided as Additional files 2–12 (see Additional files). Supplementary Fig. S1: Evaluation overview across 23 solid tumor types (cell line counts, tested pairs, and per-lineage DD AUROC values, colored by AUROC tier). Supplementary Fig. S2: Evaluation robustness: bootstrap distribution, label-permutation negative control, and external check against mutation-agnostic in4mer gold standards. Supplementary Fig. S3: CNV independence analysis. Supplementary Fig. S4: CPTAC per-cohort scatter plots. Supplementary Fig. S5: Full mutation-type stratification. Supplementary Fig. S6: Mutational co-occurrence of paralog pairs with and without TMB adjustment, pan-cancer and per-lineage. Supplementary Fig. S7: PRISM drug selectivity heatmap of the top associations passing the discovery-stage threshold. Supplementary Fig. S8: Dependency-window score (DWS) full classification. Supplementary Fig. S9: Exploratory sequence- and structure-derived descriptors: k -mer Jaccard vs. Needleman-Wunsch alignment identity validation ($n = 50$ pairs, Pearson r shown), structure-based druggability assessment of 15 proteins using ESMFold-predicted structures, feature-space structural similarity, domain conservation, and composite prioritization score. Supplementary Fig. S10: Paralog expression and overall survival in TCGA BRCA (forest plot, multivariable Cox models). Supplementary Table S1: Cell line counts per cancer type. Supplementary Table S2: Complete paralog-SL pair analysis results (110 driver $\times$ paralog $\times$ lineage associations in the three evaluation lineages (Ovarian, Endometrial, Cervical) passing the minimum sample size filter of ≥ 5 mutant and ≥ 5 wild-type cell lines). Supplementary Table S3: Curated paralog-SL pair evidence with tier assignment. Supplementary Table S4: Complete 40-gene driver panel with cancer types, selection source, and functional category. Supplementary Table S5: All 21 paralog pairs ranked by signed dependency window score with numerator, denominator, bootstrap confidence intervals, and bootstrap rank intervals. Supplementary Table S6: Gene-class-specific driver-mutation rules and per-gene variant classification (TSG: LikelyLoF; oncogene: Hotspot). Supplementary Table S7: Per-association effect sizes (Cohen's d , Hedges' g) and Welch t -test p -values for all 110 evaluated

associations (TSV file). Supplementary Table S8: TCGA BRCA overall-survival associations for the 32 paralog genes (univariable and multivariable Cox models, FDR, PH tests). Supplementary Table S9: Top paralog-SL candidates ranked by the exploratory composite prioritization score. Supplementary Table S10: Sensitivity of dependency-window score rankings and selectivity classification. Supplementary Table S11: Full regression model table for confounder controls (base, CNV-, expression-, and lineage-adjusted models with robust standard errors and BH q -values). Supplementary Table S12: Contextual literature values under non-comparable evaluation settings (published CV3 AUROCs from Feng et al. with the DD value from this study; evaluation frameworks differ and direct comparison is not warranted).

Abbreviations

SL: synthetic lethality; DD: Delta Dependency; PCS: paralog compensation score; DWS: dependency window score; TSG: tumor suppressor gene; LoF: loss of function; CPTAC: Clinical Proteomic Tumor Analysis Consortium; TCGA: The Cancer Genome Atlas; PRISM: Profiling Relative Inhibition Simultaneously in Mixtures; DepMap: Cancer Dependency Map; AUROC: area under the receiver operating characteristic curve; AUPRC: area under the precision-recall curve; MSI-H: microsatellite instability-high; MSS: microsatellite stable; MMR: mismatch repair; CNV: copy number variation; HNSCC: head and neck squamous cell carcinoma; SCLC: small cell lung cancer; NSCLC: non-small cell lung cancer; OR: odds ratio; HR: hazard ratio; CI: confidence interval; LR: logistic regression; RF: random forest; SVM: support vector machine.

Declarations

Ethics approval and consent to participate

No institutional ethics approval was required: this study analysed exclusively publicly available, de-identified data (DepMap, CPTAC, TCGA, PRISM).

Consent for publication

Not applicable.

Data availability

Raw third-party datasets. DepMap 26Q1 from the DepMap Portal [36] (exact release file-names, sizes, SHA256 checksums, and per-file download dates in the repository data manifest, `data/README.md`); CPTAC proteomics from cBioPortal [29, 30] and the Proteomic Data Commons [38]; TCGA PanCan from UCSC Xena [39]; PRISM from the DepMap PRISM portal [35]; HGNC gene families from the HGNC repository [37].

Processed analysis-ready data. Cell line-level DD values and all derived analysis tables are archived at Zenodo (concept DOI <https://doi.org/10.5281/zenodo.21502030>, which resolves to the latest release; the submission snapshot is release v1.3.2 with version DOI <https://doi.org/10.5281/zenodo.21634700>) and in the GitHub repository below.

Machine-readable supplementary tables. TSV mirrors of Supplementary Tables S1–S12 are provided as Additional files 2–12, each documented in Additional file 1.

Code. See Code availability below.

Archived manuscript-specific release. The submission snapshot is GitHub tag v1.3.2 (commit 66fc633) with version DOI <https://doi.org/10.5281/zenodo.21634700> under the concept DOI above; the accepted-version snapshot will be deposited identically.

Code availability

All analysis code is available under the MIT license at <https://github.com/tjogzt/paralogSL> (DOI: [10.5281/zenodo.21502113](https://doi.org/10.5281/zenodo.21502113)). The complete analysis pipeline (DOI: [10.5281/zenodo.21502030](https://doi.org/10.5281/zenodo.21502030)) is available at <https://github.com/tjogzt/paralog-sl-predictor>. The `paralogSL` R package (v1.1.2, 20 exported functions) provides DD computation, benchmark comparison, dependency-window scoring, CPTAC visualization, and candidate-prioritization utilities (`visualize_dependency_shift` and related helpers). Random seeds (`set.seed(42)` in R, `random_state=42` in Python) are fixed throughout.

Workflow: DepMap gene-effect matrix $\rightarrow$ `build_mutation_matrix()` $\rightarrow$ `compute_dd()` per driver $\times$ paralog $\times$ lineage $\rightarrow$ `compute_auroc()` against the curated positive set $\rightarrow$ `compute_therapeutic_window` (historical function name retained in the package; computes the DWS) for translational ranking $\rightarrow$ visualization via `plot_pancancer_summary()` and `plot_therapeutic_window()`.

Quick start (3 lines):

```
remotes::install_github("tjogzt/paralogSL") # source: github.com/tjogzt/paralogSL
```

```
1003 library(paralogSL)
1004 result <- run_paralog_analysis(dep_matrix, mut_matrix, driver="ARID1A", paralog="ARID1B")
```

1005 The package includes built-in datasets (`known_sl_pairs`, `gyn_drivers`, `solid_tumor_summary`),
1006 a vignette with worked examples, and 20 exported functions. A frozen release with all processed
1007 data tables is archived at Zenodo (concept DOI <https://doi.org/10.5281/zenodo.21502030>;
1008 submission snapshot: GitHub tag `v1.3.2`, commit `66fc633`, version DOI [https://doi.org/](https://doi.org/10.5281/zenodo.21634700)
1009 [10.5281/zenodo.21634700](https://doi.org/10.5281/zenodo.21634700)). The R package is archived separately (concept DOI [https://](https://doi.org/10.5281/zenodo.21502113)
1010 doi.org/10.5281/zenodo.21502113; v1.1.2 version DOI [https://doi.org/](https://doi.org/10.5281/zenodo.21634707)
1011 [21634707](https://doi.org/10.5281/zenodo.21634707)). Each new release mints a new version DOI under the same concept DOI, and
1012 the accepted-version snapshot will be deposited identically. The complete Python analysis
1013 pipeline (`main.py`, `pcs.py`, `pancancer.py`) is included in the same repository for full repro-
1014 ducibility. In addition, four audit scripts (`compute_headline_metrics.py`, `ml_benchmark.py`,
1015 `regression_controls.py`, and `audit_manuscript_numbers.py`) recompute every quantitative
1016 claim in this manuscript directly from raw DepMap data and derived artifacts: headline met-
1017 rics, classifier benchmarks, covariate-adjusted regressions, and all remaining numbers (lineage-
1018 level AUROCs, mechanism contrasts, MSI and mutation-type stratifications, direction audit,
1019 therapeutic-window module, pair-clustered bootstrap and permutation uncertainty estimates,
1020 power analysis, composite fold-internal scaling, composite-weight and in4mer seed sensitivi-
1021 ties, the external digenic-screen hold-out against the Harle 2025 and Flister 2025 combinato-
1022 rial screens, CPTAC RNA-layer and purity-adjusted co-variation, the TCGA UCEC/OV sur-
1023 vival extension, and machine-readable table mirrors), 420 claims in total. Each script ends
1024 with an automated claim-by-claim comparison against the reported values; a single entry point
1025 (`verify_all.sh`) runs all four audits plus the test suite in ~ 30 s on a standard workstation.

1026 Competing interests

1027 The authors declare no competing interests.

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Authors' contributions

Q.Q.M. and T.Z. conceived the study. T.Z. developed the methodology, performed all computational analyses, and developed the R package. Q.Q.M. curated clinical datasets and provided gynecological oncology domain expertise. Q.Q.M. and T.Z. wrote the manuscript.

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Additional files

Additional file 1: Supplementary Information (PDF). Supplementary Methods, Supplementary Figures S1–S10, and Supplementary Tables S1–S12.

Additional file 2: `TableS1_CellLineCounts.tsv`. Cell line counts per cancer type (23 solid tumor types).

Additional file 3: `TableS2_FullResults.tsv`. Complete paralog-SL pair analysis results (110 driver $\times$ paralog $\times$ cancer-type associations $\times$ 17 columns).

Additional file 4: `TableS3_GoldStandard.tsv`. Machine-readable mirror of the 12 evidence-tiered curated pairs (file name retained for continuity).

Additional file 5: `TableS4_DriverGenes.tsv`. Complete 40-gene driver panel with cancer types, selection source, and functional category.

Additional file 6: `TableS5_DWS.tsv`. All 21 paralog pairs ranked by signed dependency window score with numerator, denominator, floor flag, and bootstrap 95% confidence intervals.

Additional file 7: `TableS6_MutationRules.tsv`. Gene-class-specific driver-mutation rules and per-gene variant classification (TSG: LikelyLoF; oncogene: Hotspot).

Additional file 8: `TableS7_EffectSizes.tsv`. Per-association effect sizes (Cohen's d , Hedges' g) and Welch t -test p -values for all 110 evaluated associations.

Additional file 9: `TableS8_BRCA_Survival.tsv`. TCGA BRCA overall-survival associations for the 32 paralog genes (univariable and multivariable Cox models, FDR, PH tests).

Additional file 10: `TableS9_CompositeScore.tsv`. Paralog-SL candidates ranked by the exploratory composite prioritization score, annotated with the curated evidence tier of each pair (Table S3).

Additional file 11: `TableS10_DWS_Sensitivity.tsv`. Sensitivity of dependency-window score rankings and selectivity classification.

Additional file 12: `TableS11_RegressionModels.tsv`. Full regression model table for confounder controls: base, CNV-adjusted, expression-adjusted, and lineage-adjusted models per driver-paralog pair, with mutation coefficients, HC3 robust standard errors, 95% confidence intervals, nominal p -values, and Benjamini-Hochberg q -values.

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